

HazCom Plan Checklist and Annual Review

WORK AREA LOCATION AND RESPONSIBLE PARTY

In accordance with Virginia Tech Written Hazard Communications Program (available [on line on the VT EHS Web page](#)), this HazCom Plan has been prepared for:

Name of Department:	Bradley Department of Electrical and Computer Engineering
PI/Supervisor:	Richard Johnston / Jijun Niu
Building/Room #(s):	Whittemore Hall 234 A/B/C

HAZCOM COORDINATOR

☒ A Hazard Communication Coordinator or designee has been identified for this workspace.

Name:	Richard Johnston
Phone Number:	540 231 1623 / 540 320 6331
E-mail:	rickj@vt.edu
Office Location:	463 Whittemore

HAZCOM PLAN

The HazCom Plan consists of a (1) Chemical Inventory and (2) Safety Data Sheets in addition to the following area-specific information related to chemical hazards and emergency response.

A hard copy of this completed workplace-specific documentation must be maintained in a binder or folder in a designated location with Safety Data Sheets and the Chemical Inventory, where it can be accessed quickly in an emergency and/or used for lab-specific training.

☒ *Hazardous Chemicals Inventory/ List* for this workplace has been prepared. The Hazard Communications Coordinator is responsible for updating and managing the list.

Location of Chemical Inventory/List:	The white binder is mounted on the storage room door located to the right of the main entrance
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☒ *Safety Data Sheets* for all chemicals (current versions) have been compiled.

Primary SDS Location:	The white binder is mounted on the storage room door located to
Back-up SDS Location:	On the top shelf of the metal locker in the storage room

Note: Safety data sheets (SDSs) must be readily accessible (no barriers to student or employee access) to personnel working in the lab. In addition to the primary location of SDSs, a back-up SDS management system must be available in the event of an emergency (including power outages, equipment failure, on-line access delays, etc.)

****Important:** When possible, always provide responders (local hospital and rescue squad) with hard copies of SDS(s) as it will greatly assist them in responding quickly to your lab's emergency.**

HazCom Plan Checklist and Annual Review

NON-ROUTINE TASKS

- ☒ Non-routine tasks and their hazards have been identified, and controls to minimize exposure are described below.

PERSONNEL TRAINING

- ☒ Personnel have been informed about the labeling system(s) used in this space to communicate chemical hazards (NFPA, HMIS or other).

Unless otherwise specified below, the chemical/product name and any primary hazard word will be marked on secondary container.

- ☒ All personnel using work-area chemicals/chemical products have completed on-line training and received workplace-specific training from the HazCom coordinator.
- ☒ All personnel are entered in the Safety Management System for this work area, and their training has been recorded.

ANNUAL UPDATE/ REVISIONS/ REVIEW

- ☒ Review the HazCom Plan contents annually. Record the annual review below.

Date	Revision #	Comments	Signature
6/5/2026	V1	All good	J.N.

Chemical List

Chemical Name (as it appears on the container and SDS)	Location or Responsible Person
1. Isopropyl Alcohol 99%	234B
2. PLA	234C 3D printer
3. PETG	234C 3D printer
4. Sn99 Lead Free Solder	234C soldering station
5. 275 Sn96.5Ag3.0Cu0.5 Solder Wire	234B
6. LiFePO4 Rechargeable Battery	234A ammo crater
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Safety Data Sheet Isopropyl Alcohol 99%

SDS Revision Date: 07/17/2023

Section 1. Identification

1.1. Product identifier

Product Identity

Isopropyl Alcohol 99%

Alternate Names

Isopropyl Alcohol 99%, Formula No.: 005

1.2. Relevant identified uses of the substance or mixture and uses advised against

Intended use

See Technical Data Sheet.

1.3. Details of the supplier of the safety data sheet

Company Name

Hydrox Laboratories
825 Tollgate Rd.
Elgin, IL 60123

Emergency

24 hour Emergency Telephone No.

800-255-3924

Customer Service: Hydrox Laboratories

847-468-9400

Section 2. Hazard(s) identification

2.1. Classification of the substance or mixture

Flam. Liq. 2;H225

Highly Flammable liquid and vapor.

Eye Dam. 2A;H319

Causes serious eye irritation.

STOT SE 3;H336

May cause drowsiness or dizziness.

2.2. Label elements

(Not required on cosmetic product or case labels per Occupational Safety and Health Standards 29 CFR 1910.1200(b)(5))



Danger

H225 Highly flammable liquid and vapor.

H319 Causes serious eye irritation.

H336 May cause drowsiness and dizziness.

[Prevention]

P210 Keep away from heat, sparks, open flames, and other ignition sources - No smoking.

P233 Keep container tightly closed.

P240 Ground, bond container and receiving equipment.



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P241 Use explosion-proof electrical, ventilating, light, equipment.

P242 Use only non-sparking tools.

P243 Take precautionary measures against static discharge.

P261 Avoid breathing dust, fume, gas, mist, vapors, spray.

P264 Wash thoroughly after handling.

P271 Use only outdoors or in a well-ventilated area.

P280 Wear protective gloves, eye protection, face protection.

[Response]

P303+361+353 IF ON SKIN (or hair): Remove, take off immediately all contaminated clothing. Rinse skin with water, shower.

P304+340 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.

P305+351+338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P312 Call a POISON CENTER, doctor or physician if you feel unwell.

P337+313 If eye irritation persists: Get medical advice or attention.

P370+378 In case of fire: Use extinguishing media listed in section 5 of SDS for extinction.

[Storage]

P403+233 Store in a well ventilated place. Keep container tightly closed.

P403+235 Store in a well ventilated place. Keep cool.

P405 Store locked up.

[Disposal]

P501 Dispose of contents or container in accordance with local and national regulations.

Section 3. Composition/information on ingredients

This product contains the following substances that present a hazard within the meaning of the relevant State and Federal Hazardous Substances regulations.

Ingredient/Chemical Designations	Weight %	GHS Classification	Notes
Isopropyl Alcohol CAS Number: 0000067-63-0	75 - 100	Flam. Liq. 2;H225 Eye Irrit. 2;H319 STOT SE 3;H336	----

The specific chemical identity and/or exact percentage (concentration) of composition has been withheld as a trade secret.

*PBT/vPvB - PBT-substance or vPvB-substance.

The full texts of the phrases are shown in Section 16.

Section 4. First aid measures

4.1. Description of first aid measures

General

In all cases of doubt, or when symptoms persist, seek medical attention.
Never give anything by mouth to an unconscious person.



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Inhalation	Remove to fresh air, keep patient warm and at rest. If breathing is irregular or stopped, give artificial respiration. If unconscious, place in the recovery position and obtain immediate medical attention. Give nothing by mouth.
Eyes	Irrigate copiously with clean water for at least 15 minutes, holding the eyelids apart and seek medical attention.
Skin	Remove contaminated clothing. Wash skin thoroughly with soap and water or use a recognized skin cleanser.
Ingestion	If swallowed obtain immediate medical attention. Keep at rest. Do NOT induce vomiting.

4.2. Most important symptoms and effects, both acute and delayed

Overview	Signs and Symptoms of Exposure: Giddiness, headache, dizziness and nausea. Health Hazards (Acute and Chronic): Generally used as a rubdown. Vapor irritates eyes. High concentration of vapor can irritate respiratory tract, is anesthetic and may cause CNS depression. Medical Conditions Generally Aggravated by Exposure: Pre-existing and respiratory disorders, may be aggravated by exposure. Treat symptomatically. Exposure to solvent vapor concentrations from the component solvents in excess of the stated occupational exposure limits may result in adverse health effects such as mucous membrane and respiratory system irritation and adverse effects on the kidneys, liver and central nervous system. Symptoms include headache, nausea, dizziness, fatigue, muscular weakness, drowsiness and in extreme cases, loss of consciousness. Repeated or prolonged contact with the preparation may cause removal of natural fat from the skin resulting in dryness, irritation, and possible non-allergic contact dermatitis. Solvents may also be absorbed through the skin. Splashes of liquid in the eyes may cause irritation and soreness with possible reversible damage. Check section 2.2 (GHS Label Elements) for further details.
Inhalation	May cause drowsiness or dizziness.
Eyes	Causes serious eye irritation.

Section 5. Fire-fighting measures

5.1. Extinguishing media

Recommended extinguishing media; alcohol resistant foam, CO², powder, water spray.
Unsuitable extinguishing media: Do not use; water jet.

5.2. Special hazards arising from the substance or mixture

Hazardous decomposition: No hazardous decomposition data available.
Keep away from heat, sparks, open flames, and other ignition sources - No smoking.
Keep container tightly closed.
Ground, bond container and receiving equipment.
Use explosion-proof electrical, ventilating, light, equipment.
Use only non-sparking tools.



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Take precautionary measures against static discharge.

Avoid breathing dust, fume, gas, mist, vapors, spray.

5.3. Advice for fire-fighters

As with all fires, wear positive pressure, self-contained breathing apparatus, (SCBA) with a full face piece and protective clothing. Persons without respiratory protection should leave area. Wear SCBA during clean-up immediately after fire. No smoking.

Dilution of burning liquid with water will affect extinguishment.

ERG Guide No. 129

Section 6. Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

Put on appropriate personal protective equipment (see section 8).

6.2. Environmental precautions

Do not allow spills to enter drains or waterways.

Use good personal hygiene practices. Wash hands before eating, drinking, smoking, or using toilet. Promptly remove soiled clothing and wash thoroughly before reuse.

6.3. Methods and material for containment and cleaning up

Eliminate all sources of ignition. Small spills should be flushed with large quantities of water. Larger spills should be collected for disposal.

Section 7. Handling and storage

7.1. Precautions for safe handling

Handle containers carefully to prevent damage and spillage.

Do not take internally. Flammable liquid.

Check section 2.2 (GHS Label Elements) for further details. - [Prevention]

7.2. Conditions for safe storage, including any incompatibilities

Incompatible materials: No data available.

Keep away from heat, sparks and open flames. Keep container closed.

Check section 2.2 (GHS Label Elements) for further details. - [Storage]

7.3. Specific end use(s)

No data available.



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Section 8. Exposure controls / personal protection

8.1. Control parameters

CAS No.	Ingredient	Exposure	
		Source	Value
0000067-63-0	Isopropyl Alcohol	OSHA	TWA 400 ppm (980 mg/m ³) STEL 500 ppm
		ACGIH	TWA: 200 ppm STEL: 400 ppm
		NIOSH	TWA 400 ppm (980 mg/m ³) STEL: 500 ppm (1225 mg/m ³)

8.2. Exposure controls

Respiratory	If workers are exposed to concentrations above the exposure limit, they must use the appropriate, certified respirators.
Eyes	Chemical resistant goggles
Skin	Protective gloves recommended.
Engineering Controls	Provide adequate ventilation. Where reasonably practicable this should be achieved by the use of local exhaust ventilation and good general extraction. If these are not sufficient to maintain concentrations of particulates and any vapor below occupational exposure limits suitable respiratory protection must be worn.
Other Work Practices	Use good personal hygiene practices. Wash hands before eating, drinking, smoking, or using toilet. Promptly remove soiled clothing and wash thoroughly before reuse.

Check section 2.2 (GHS Label Elements) for further details.

Section 9. Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance	Color: Colorless Physical State: Liquid
Odor	Characteristic
Odor threshold	Not determined
pH	Not Measured
Melting point / freezing point	-89°C (-128°F)
Initial boiling point and boiling range	Not Measured
Flash Point	12°C (54°F) [ASTM D-56]
Evaporation rate (Ether = 1)	3.9 [in-house method]
Flammability (solid, gas)	Not Applicable
Upper/lower flammability or explosive limits	Lower Explosive Limit: 2.0 Upper Explosive Limit: 12.0
Vapor pressure (Pa)	4.3 kPa (32.25mm Hg) at 20°C [Calculated] [In-house method]



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Vapor Density	>1 at 101 kPa [Calculated]
Relative Density	0.800 - 0.833 @ 20°C
Solubility in Water	Complete
Partition coefficient n-octanol/water (Log Kow)	Not Measured
Auto-ignition temperature	>350°C (662°F) [Technical literature]
Decomposition temperature	Not Measured
Viscosity (cSt)	Not Measured
Isopropyl Alcohol Assay	99.5 – 100%

9.2. Other information

No other relevant information.

Section 10. Stability and reactivity

10.1. Reactivity

Hazardous Polymerization will not occur.

10.2. Chemical stability

Stable under normal circumstances.

10.3. Possibility of hazardous reactions

No data available.

10.4. Conditions to avoid

No data available.

10.5. Incompatible materials

No data available.

10.6. Hazardous decomposition products

No hazardous decomposition data available.

Section 11. Toxicological information

Acute toxicity

Exposure to solvent vapor concentrations from the component solvents in excess of the stated occupational exposure limits may result in adverse health effects such as mucous membrane and respiratory system irritation and adverse effects on the kidneys, liver and central nervous system. Symptoms include headache, nausea, dizziness, fatigue, muscular weakness, drowsiness and in extreme cases, loss of consciousness.

Repeated or prolonged contact with the preparation may cause removal of natural fat from the skin resulting in dryness, irritation, and possible non-allergic contact dermatitis. Solvents may also be absorbed through the skin. Splashes of liquid in the eyes may cause irritation and soreness with possible reversible damage.

Note: When no route specific LD50 data is available for an acute toxin, the converted acute toxicity point estimate was used in the calculation of the product's ATE (Acute Toxicity Estimate).



Safety Data Sheet

Isopropyl Alcohol 99%

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Ingredient	Oral LD50, mg/kg	Skin LD50, mg/kg	Inhalation Vapor LC50, mg/L/4hr	Inhalation Dust/Mist LC50, mg/L/4hr	Inhalation Gas LC50, ppm
Isopropyl Alcohol - (67-63-0)	5,840.00, Rat - Category: NA	16,400.00, Rabbit - Category: NA	72.60, Rat - Category: NA	No data available.	No data available.

Carcinogen Data

CAS No.	Ingredient	Source	Value
0000067-63-0	Isopropyl Alcohol	OSHA	Regulated Carcinogen: No;
		NTP	Known: No; Suspected: No;
		IARC	Group 1: No; Group 2a: No; Group 2b: No; Group 3: Yes; Group 4: No;
		ACGIH	A4

Classification	Category	Hazard Description
Acute toxicity (oral)	---	Not Applicable
Acute toxicity (dermal)	---	Not Applicable
Acute toxicity (inhalation)	---	Not Applicable
Skin corrosion/irritation	---	Not Applicable
Serious eye damage/irritation	2A	Causes serious eye irritation.
Respiratory sensitization	---	Not Applicable
Skin sensitization	---	Not Applicable
Germ cell mutagenicity	---	Not Applicable
Carcinogenicity	---	Not Applicable
Reproductive toxicity	---	Not Applicable
STOT-single exposure	---	Not Applicable
STOT-single exposure	3	May cause drowsiness or dizziness.
STOT-repeated exposure	---	Not Applicable
Aspiration hazard	---	Not Applicable

Section 12. Ecological information

12.1. Toxicity

No additional information provided for this product. See Section 3 for chemical specific data.

Aquatic Ecotoxicity

Ingredient	96 hr LC50 fish, mg/L	48 hr EC50 crustacea, mg/L	ErC50 algae, mg/L
Isopropyl Alcohol - (67-63-0)	10,000.00, Pimephales promelas	No data available.	No data available.



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12.2. Persistence and degradability

There is no data available on the preparation itself.

12.3. Bioaccumulative potential

Not Measured

12.4. Mobility in soil

No data available.

12.5. Results of PBT and vPvB assessment

This product contains no PBT/vPvB chemicals.

12.6. Other adverse effects

No data available.

Section 13. Disposal considerations

13.1. Waste treatment methods

Observe all federal, state, and local regulations when disposing of this substance.

Section 14. Transport information

	DOT (Domestic Surface Transportation)	IMO / IMDG (Ocean Transportation)	ICAO/IATA
14.1. UN number	UN1219	UN1219	UN1219
14.2. UN proper shipping name	UN1219, Isopropanol or Isopropyl alcohol, 3, II	Isopropanol or Isopropyl alcohol	Isopropanol or Isopropyl alcohol
14.3. Transport hazard class(es)	DOT Hazard Class: 3 Sub Class: Not Applicable	IMDG: 3 Sub Class: Not Applicable	Air Class: 3 Sub Class: Not Applicable
14.4. Packing group	II	II	II
14.5. Environmental hazards	Marine Pollutant: No;		
14.6. Special precautions for user	Not Applicable		

Section 15. Regulatory information

Regulatory Overview	The regulatory data in Section 15 is not intended to be all-inclusive, only selected regulations are represented.
Toxic Substance Control Act (TSCA)	All components of this material are either listed or exempt from listing on the TSCA Inventory.
EPCRA 302 Extremely Hazardous:	To the best of our knowledge, there are no chemicals at levels which require reporting under this statute.



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EPCRA 313 Toxic Chemicals:

Isopropyl Alcohol

Proposition 65 - Carcinogens (>0.0%):

To the best of our knowledge, there are no chemicals at levels which require reporting under this statute.

Proposition 65 - Developmental Toxins (>0.0%):

To the best of our knowledge, there are no chemicals at levels which require reporting under this statute.

Proposition 65 - Female Repro Toxins (>0.0%):

To the best of our knowledge, there are no chemicals at levels which require reporting under this statute.

Proposition 65 - Male Repro Toxins (>0.0%):

To the best of our knowledge, there are no chemicals at levels which require reporting under this statute.

Proposition 65 Label Warning:

This product contains no chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

Section 16. Other information

Revision Date 07/17/2023

The information and recommendations contained herein are based upon data believed to be correct. However, no guarantee or warranty of any kind, expressed or implied, is made with respect to the information contained herein. We accept no responsibility and disclaim all liability for any harmful effects which may be caused by exposure to our products. Customers/users of this product must comply with all applicable health and safety laws, regulations, and orders.

The full text of the phrases appearing in section 3 is:

H225 Highly flammable liquid and vapor.

H319 Causes serious eye irritation.

H336 May cause drowsiness and dizziness.

Disclaimer: The contents of this SDS are believed to be correct but do not purport to be all-inclusive and should only be used as a guide. Hydrox Laboratories, Inc. disclaims any express or implied warranty as to the accuracy of the above information and shall not be held liable for any direct, incidental or consequential damages resulting from the reliance on the above information.

End of Document

SN 99 LEAD FREE SOLDER

4901

Safety Data Sheet

Section 1: Identification

Product Identifier and Other Means of Identification

Product Name: Sn99 Lead Free Solder**SDS Code:** 4901**Related Part #** 4901-112G, 4901-227G, 4901-454G, 4901-2LB

Recommended Use and Restriction on Use

Use: Lead free solder wire**Uses Advised Against:** Do not use brazing soldering methods such as high temperature torch soldering/torch welding.

Details of Manufacturer or Importer

ManufacturerMG Chemicals
1210 Corporate Drive
Burlington, Ontario L7L 5R6
CANADAMG Chemicals (Head Office)
9347-193 Street
Surrey, British Columbia V4N 4E7
CANADA**☎** +1-800-340-0772
FAX +1-800-340-0773
E-MAIL support@mgchemicals.com
WEB www.mgchemicals.com**☎** +1-905-331-1396
FAX +1-905-331-2682
E-MAIL info@mgchemicals.com**E-MAIL** (Competent Person): sds@mgchemicals.com

Emergency Phone Number

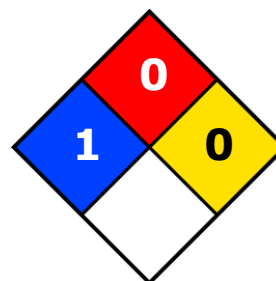
For hazardous material incidents ONLY—leaks, spills, fires, exposures or accidents
USA or CANADA: Call CHEMTREC ☎: **+1-800-424-9300****For emergencies involving dangerous goods;** Collect 24/7
CANADA: Call CANUTEC ☎: **+1-613-996-6666** or ***666** on cellular phones

SN 99 LEAD FREE SOLDER
4901
Section 2: Hazard(s) Identification
Classification of Hazardous Chemical
GHS Categories

Based on available data, this product does not meet the HCS 2012 or WHMIS 2015 classification criteria.

Other Classifications
HMIS® RATING

HEALTH:	1
FLAMMABILITY:	0
PHYSICAL HAZARD:	0
PERSONAL PROTECTION:	

NFPA® 704 CODES


Approximate HMIS and NFPA Risk Ratings Legend:

0 (Low or none); 1 (Slight); 2 (Moderate); 3 (Serious); 4 (Severe)

Label Elements

Signal Word	<i>None Mandated</i>
Pictograms	Hazard Statements
<i>None mandated</i>	None

Hazards Not Otherwise Classified

Not applicable

SN 99 LEAD FREE SOLDER**4901****Section 3: Composition/Information on Ingredients**

CAS #	Chemical Name	%(wt/wt)
7440-31-5	tin	97%
7440-50-8	copper	0.5%

Note: Also contains 2.2% non hazardous, hydrogenated rosin flux

Section 4: First-Aid Measures

<i>Exposure Condition</i>	<i>GHS Code/Symptoms/Precautionary Statements</i>
IF INHALED	P304 + P340
Immediate Symptoms	<i>cough, irritation of the respiratory track</i>
Response	Remove person to fresh air and keep comfortable for breathing.
IF IN EYES	P305 + P351 + P338, P337 + P313
Immediate Symptoms	<i>redness, mild irritation</i>
Response	Rinse cautiously with water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention.
IF ON SKIN	P302 + P352
Immediate Symptoms	<i>mild irritation</i>
Response	Wash with plenty of water.
IF SWALLOWED	P301 + P330, P331
Immediate Symptoms	None known
Response	Rinse mouth. Do NOT induce vomiting.

SN 99 LEAD FREE SOLDER**4901****Section 5: Fire-Fighting Measures**

In case of fire	P370 + P378
Extinguishing Media	Use extinguish media suitable for surrounding.
Specific Hazards	In a fire, this product can release metal oxide fumes and irritation flux fumes. Do NOT use water on fires where molten metal is present.
Combustion Products	Produces CO and CO ₂ , and tin oxides (SnO _x).
Fire-Fighter	Wear self-contained breathing apparatus and full fire-fighting turn-out gear.

Section 6: Accidental Release Measures

Personal Protection	See personal protection recommendations in Section 8.
Precautions for Response	Avoid breathing the fumes. Remove or keep away all sources of extreme heat.
Environmental Precautions	Avoid releasing to the environment.
Containment Methods	Not applicable
Cleaning Methods	Collect waste in a sealable waste container. Reuse molten material if it is not contaminated.
Disposal Methods	Dispose of spill waste according to Section 13.

SN 99 LEAD FREE SOLDER**4901****Section 7: Handling and Storage**

- Prevention** Keep out of reach of children.
Avoid breathing fumes.
Do not eat, drink, or smoke when using this product.
- Handling** Wear protective gloves/clothing/eye protection.
Wash hands thoroughly after handling.
Avoid release to the environment.
- Storage** Not applicable.

Section 8: Exposure Controls/Personal Protection**Routes of Entry**

Inhalation, Ingestion, and Eye Contact

Substances with Occupational Exposure Limit Values

Chemical Name	Country	Long Term Exposure Limits (PEL)	Short Term Exposure Limits (STEL)
tin	ACGIH	2 mg/m ³	Not established
	U.S.A. OSHA PEL	2 mg/m ³	Not established
	Canada AB	2 mg/m ³	Not established
	Canada BC	2 mg/m ³	Not established
	Canada ON	2 mg/m ³	Not established
	Canada QC	2 mg/m ³	Not established

Section continued on the next page

SN 99 LEAD FREE SOLDER**4901***Continued...*

Chemical Name	Country	Long Term Exposure Limits (PEL)	Short Term Exposure Limits (STEL)
copper (dust and mist)	ACGIH	1.0 mg/m ³	Not established
	U.S.A. OSHA PEL	1.0 mg/m ³	Not established
	Canada AB	1.0 mg/m ³	Not established
	Canada BC	1.0 mg/m ³	Not established
	Canada ON	1 mg/m ³	Not established
	Canada QC	1 mg/m ³	Not established

Note: Ingredients are listed in descending weight contribution order (from greatest to least). The ACGIH², OSHA (Table Z-1), and Canadian provinces exposure limits were consulted. Limits from by RTECS database¹ of the Canadian Centre for Occupational Health and Safety (CCOHS) a data from suppliers' SDS were also consulted. Short term exposure limits (STEL) are for 15 min and long term permissible exposure limits (PEL) for 8 h.

Engineering Controls**Ventilation**

Keep airborne concentrations below exposure limits.

Manufacturer's Note: Because soft soldering temperatures are generally too low to generate metal vapors, fumes or dust, the risks of metal or metal compound generation are negligible. However, the use of a local exhaust system is highly recommended.

The iron soldering temperatures are high enough to generate potentially toxic fumes due to the volatilization or degradation of the flux and of the coating material on the soldered surface.

Personal Protective Equipment**Eye protection**

Wear appropriate protective eyeglasses or chemical safety goggles.

Recommendation: Ensure that glasses have side shields for lateral protection.

Skin Protection

For incidental contacts, use nitrile or other chemically resistant gloves. Thermal resistant gloves should be worn instead if contact with molten metal is expected.

Section continued on the next page

SN 99 LEAD FREE SOLDER**4901**

Respiratory Protection If exposed to fumes or dust above the exposure limit, a suitable wear respirator meeting local/regional/national guidelines.

Generally, for emergencies and exposure above 0.01 mg/m³, use a self-contained breathing apparatus with full face piece operated in a pressure positive mode.

RECOMMENDATION: Consult your local safety supply store to ensure your respirator has filter cartridges appropriate for the ingredients listed in section 3 of this SDS, and that the respirator is fitted to the employee by a professional.

General Hygiene Considerations

Wash hands thoroughly with water and soap after handling.

Section 9: Physical and Chemical Properties

Physical State	Solid	Lower Flammability Limit	Not applicable
Appearance	Silver grey	Upper Flammability Limit	Not applicable
Odor	None	Vapor Pressure @20 °C	Not available
Odor Threshold	Not available	Vapor Density	Not applicable
pH	Not available	Specific Gravity @25 °C	7.4
Freezing/Melting Point	228 °C [442 °F]	Solubility in Water	Negligible ^{a)}
Boiling Point	Not available	Partition Coefficient	Not available
Flash Point	Not applicable	Auto-ignition Temperature	Not available
Evaporation Rate	Not available	Decomposition Temperature	Not available
Flammability (solid, gas)	Not applicable	Viscosity @25 °C	Not applicable

a) Metal components are sparingly soluble

SN 99 LEAD FREE SOLDER**4901****Section 10: Stability and Reactivity**

Reactivity	Not available
Chemical Stability	Chemically stable at normal temperatures and pressures
Conditions to Avoid	Extreme temperatures above 450 °C, such as those due to welding
Incompatibilities	Oxidizing agents, strong acids
Polymerization	Will not occur
Decomposition	Will not decompose under normal conditions. For thermal decomposition, see combustion products in Section 5.

Section 11: Toxicological Information**Routes of Exposure**

Inhalation, Ingestion, and Eye Contact

Symptoms Summary

Eyes	May cause redness, mild irritation
Skin	May cause mild irritation.
Inhalation	May cause nose, throat and lung irritation. Overexposure to dust or metal fumes may lead to pneumoconiosis (or Stannosis), anemia, central nervous system effects
Ingestion	No effect known
Chronic	No effect known

Section continued on the next page

SN 99 LEAD FREE SOLDER**4901****Acute Toxicity (Lethal Exposure Concentrations)**

Chemical Name	LD50 oral	LD50 dermal	LC50 inhalation
tin	>2 000 mg/kg Rat	>2 000 mg/kg Rabbit	4.75 mg/m ³ Rat 4 h
copper	>5 000 mg/kg Mouse	Not available	>5.11 mg/L Rat 4 h

Note: Toxicity data from the RTECS database accessed through the Canadian Centre for Occupational Health and Safety (CCOHS)² were consulted. The data from supplier (M)SDS were also consulted.

Other Toxicological Effects

Skin corrosion/irritation	Based on available data, the classification criteria are not met
Serious eye damage/irritation	Based on available data, the classification criteria are not met
Sensitization (allergic reactions)	Based on available data, the classification criteria are not met
Carcinogenicity (risk of cancer)	Not classified or listed as a carcinogen by IARC, ACGIH, CA Prop 65, or NTP
Mutagenicity (risk of heritable genetic effects)	Based on available data, the classification criteria are not met
Reproductive Toxicity (risk to sex functions)	Based on available data, the classification criteria are not met
Teratogenicity (risk of fetus malformation)	Based on available data, the classification criteria are not met
STOT-single exposure	Based on available data, the classification criteria are not met
STOT-repeated exposure	Based on available data, the classification criteria are not met
Aspiration hazard	Not applicable. This product doesn't contain any Cat 1 ingredients and has a viscosity >>20 mm/s ² .

SN 99 LEAD FREE SOLDER**4901****Section 12: Ecological Information**

The IMDG Code criteria, the raw-material safety data sheets, and supporting data from the European Chemical Agency database (<http://echa.europa.eu>) were used to support the classification.

Based on transformation/dissolution data published by ECHA registrants, the classification threshold is not met for massive copper.

Based on available data for tin and hydrogenated rosin, the GHS aqueous toxicity classification criteria are not met.

Acute Ecotoxicity

Non hazardous

Chronic Ecotoxicity

Non hazardous

Biodegradability

Not available

Bioaccumulation

Not available

Other Effects

Not available

Section 13: Disposal Information

Dispose of contents in accordance with all local, regional, national, and international regulations.

SN 99 LEAD FREE SOLDER**4901****Section 14: Transport Information****Ground**

Refer to TDG regulations (Canadian Transportation of Dangerous Goods regulations);
USA DOT 49 CFR (Parts 100 to 185) **Regulations.**

Non Regulated

Air

Refer to ICAO-IATA Dangerous Goods Regulations.

Non Regulated

Sea

Refer to IMDG regulations.

Non Regulated

Section 15: Regulatory Information**Canada****WHMIS 1988 Classification**

This mixture doesn't contain any ingredient giving rise to a classification under WHMIS.

Domestic Substance List (DSL) / Non-Domestic Substance Lists (NDSL)

All hazardous ingredients are listed on the DSL.

Industry and Science Canada

MG Labels products intended for the workplace to conform to WHMIS labeling regulations. Product identification, net quantity declaration, minimum printing type size heights, and packaging of this product are in compliance.

Health Canada

Products produced by MG Chemicals intended for retail display conform to the Canadian Consumer Labeling Regulations.

Section continued on the next page

SN 99 LEAD FREE SOLDER**4901****USA****CAA** (Clean Air Act, USA)

This product does not contain any class 1 ozone depleting substances.

This product does not contain any class 2 ozone depleting substances.

This product does not contain substances that are listed as hazardous air pollutants.

EPCRA (Emergency Planning and Right to Know Act, USA, 40 CFR 372.45)

This product does not contain ingredients that subject to the reporting requirements of section 313 Title III of the SARA of 1986 and 40 CFR part 372.

TSCA (Toxic Substances Control Act of 1976, USA)

All substances are TSCA listed.

California Proposition 65 (Chemicals known to cause cancer or reproductive toxicity, June 06, 2014 revision, USA).

This product does not contain any of the listed substances.

Europe**RoHS** (Restriction of Hazardous Substances Directive)

This product does not contain any lead, cadmium, mercury, hexavalent chromium, PBB's, or PBDE's, and complies with European RoHS regulations.

WEEE (Waste Electrical and Electronic Equipment Directive)

This product is subject to the WEEE regulation.

Section 16: Other Information

SDS Prepared by Michel Hachey

Date of Review 20 April 2015

Supersedes 03 June 2014

Reason for Changes: Change over to HCS 2012 and WHMIS 2015 requirements.

Section continued on the next page

SN 99 LEAD FREE SOLDER**4901****Reference**

- 1) ACGIH 2013 TLVs and BEIs: Based on the documentation of the threshold limit values for chemical substances and physical agents & biological exposure indices, American Conference of Governmental of Industrial Hygienist Cincinnati, OH (2013).
- 2) All toxicological data were checked against the RTECS (Registry of Toxic Effects of Chemical Substances®)

Abbreviations

ACGIH	American Conference of Governmental Industrial Hygienists (USA)
EC50	Half maximal effective concentration
EL50	Half maximal effective loading
IARC	International Agency for Research on Cancer
NOELR	No observable effect loading ratio
NTP	National Toxicology Program
GHS	Globally Harmonized System of Classification of Labeling of Chemicals
LC50	Lethal Concentration 50%
LCLo	Lowest published lethal concentration
LD50	Lethal Dose 50%
OEL	Occupational Exposure Limit
PEL	Permissible Exposure Limit
SDS	Safety Data Sheet
STEL	Short-Term Exposure Limit
TCLo	Lowest published toxic concentration
TWA	Time Weighted Average
VOC	Volatile Organic Content

Technical Queries Contact us regarding any questions, improvement suggestions, or problems with this product. Application notes, instructions, and FAQs are located at www.mgchemicals.com.

Email: support@mgchemicals.com

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Disclaimer This material safety data sheet is provided as an information resource only. M.G. Chemicals, Ltd. believes the information contained herein is accurate and compiled from reliable sources. It is the responsibility of the user to query and verify any information seeming suspect where doubt on the validity may exist. The buyer assumes all responsibility of using and handling the product in accordance with local, regional, national, and international regulations.

Safety Data Sheet

FOR REGULATORY AND SDS QUESTIONS (U.S. AND CANADA):
CALL THE PRODUCT STEWARDSHIP LINE
1- 908-791-2336 9 AM TO 6 PM ET (Mon-Fri)

Section 1. Identification

Product name : 275 Sn96.5Ag3.0Cu0.5 Solder Wire
Product code : 275Alloy771
Product type : Solid.
Date of issue/Date of revision : March 10 2023.

Manufacturer - Supplier	Telephone no.:	Emergency phone:
Alpha Assembly Solutions Inc. 800 West Thorndale Avenue Itasca, IL 60143 USA	 800-253-7837 1-630-616-4000	DOMESTIC NORTH AMERICA 202-464-2554
ALPHA METALS MEXICO SA DE CV Ave Nafta 800, Parque Industrial STIVA Apodaca NL 66600 Mexico	Tel: +52 81 1156-6602	Tel: 01 800 022 1400 Tel: +52 55 5559-1588
Alpha Assembly Solutions Brasil Soldas Ltda Rio Jaguarão, 1540 - Vila Buriti Manaus Amazonas 69072-055 Brasil	Tel: 55 92 3614-7400	Tel: 55 92 3614-7423

Section 2. Hazards identification

OSHA/HCS status : While this material is not considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200), this SDS contains valuable information critical to the safe handling and proper use of the product. This SDS should be retained and available for employees and other users of this product.

Classification of the substance or mixture : Not classified.

GHS label elements

Signal word :  No signal word.

Hazard statements : No known significant effects or critical hazards.

Precautionary statements

Prevention :  When using do not eat, drink or smoke.

Response : Get medical attention if you feel unwell.

Storage :  Keep container tightly closed in a cool, well-ventilated place.

Disposal : Dispose of contents and container in accordance with all local, regional, national and international regulations.

Hazards not otherwise classified : None known.

Section 3. Composition/information on ingredients

Substance/mixture : Mixture

Ingredient name	%	CAS number
tin	80-100	7440-31-5
silver	1-10	7440-22-4
Proprietary Rosin/Resin	1-10	-

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified and hence require reporting in this section.

Occupational exposure limits, if available, are listed in Section 8.

Section 4. First aid measures

Description of necessary first aid measures

- Eye contact** : Immediately flush eyes with plenty of water, occasionally lifting the upper and lower eyelids. Check for and remove any contact lenses. Get medical attention if irritation occurs.
- Inhalation** : Remove victim to fresh air and keep at rest in a position comfortable for breathing. Get medical attention if symptoms occur.
- Skin contact** : Flush contaminated skin with plenty of water. Remove contaminated clothing and shoes. Get medical attention if symptoms occur.
- Ingestion** : ☒ Wash out mouth with water. If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Do not induce vomiting unless directed to do so by medical personnel. Get medical attention if symptoms occur.

Most important symptoms/effects, acute and delayed

Potential acute health effects

- Eye contact** : No known significant effects or critical hazards.
- Inhalation** : No known significant effects or critical hazards.
- Skin contact** : No known significant effects or critical hazards.
- Ingestion** : No known significant effects or critical hazards.

Over-exposure signs/symptoms

- Eye contact** : No specific data.
- Inhalation** : No specific data.
- Skin contact** : No specific data.
- Ingestion** : No specific data.

Indication of immediate medical attention and special treatment needed, if necessary

- Notes to physician** : Treat symptomatically. Contact poison treatment specialist immediately if large quantities have been ingested or inhaled.
- Specific treatments** : No specific treatment.
- Protection of first-aiders** : No action shall be taken involving any personal risk or without suitable training.

See toxicological information (Section 11)

Section 5. Fire-fighting measures

Extinguishing media

Suitable extinguishing media : Use an extinguishing agent suitable for the surrounding fire.

Unsuitable extinguishing media : None known.

Specific hazards arising from the chemical : No specific fire or explosion hazard.

Hazardous thermal decomposition products : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
metal oxide/oxides

Special protective actions for fire-fighters : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training.

Special protective equipment for fire-fighters : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

Section 6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

For non-emergency personnel : No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilled material. Put on appropriate personal protective equipment.

For emergency responders : If specialized clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials. See also the information in "For non-emergency personnel".

Environmental precautions : Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

Methods and materials for containment and cleaning up

Small spill : Move containers from spill area. Vacuum or sweep up material and place in a designated, labeled waste container. Dispose of via a licensed waste disposal contractor.

Large spill : Move containers from spill area. Prevent entry into sewers, water courses, basements or confined areas. Vacuum or sweep up material and place in a designated, labeled waste container. Dispose of via a licensed waste disposal contractor. Note: see Section 1 for emergency contact information and Section 13 for waste disposal.

Section 7. Handling and storage

Precautions for safe handling

Protective measures : Put on appropriate personal protective equipment (see Section 8).

Advice on general occupational hygiene : Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.

Section 7. Handling and storage

Conditions for safe storage, including any incompatibilities : Store in accordance with local regulations. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10) and food and drink. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabeled containers. Use appropriate containment to avoid environmental contamination. See Section 10 for incompatible materials before handling or use.

Section 8. Exposure controls/personal protection

Control parameters

Occupational exposure limits



silver

ACGIH TLV (United States, 3/2017).

TWA: 2 mg/m³, (as Sn) 8 hours.

NIOSH REL (United States, 10/2016).

TWA: 2 mg/m³, (as Sn) 10 hours.

OSHA PEL (United States, 6/2016).

TWA: 2 mg/m³, (as Sn) 8 hours.

ACGIH TLV (United States, 3/2017). Notes: Substances for which the TLV is higher than the OSHA Permissible Exposure Limit (PEL) and/or the NIOSH Recommended Exposure Limit (REL).

See CFR 58(124) :36338-33351, June 30, 1993, for revised OSHA PEL.

TWA: 0.1 mg/m³ 8 hours. Form: Dust and fumes

NIOSH REL (United States, 10/2016). Notes: as Ag

TWA: 0.01 mg/m³, (as Ag) 10 hours. Form: METAL DUST AND SOLUBLE

OSHA PEL 1989 (United States, 3/1989).

TWA: 0.01 mg/m³, (as Ag) 8 hours.

OSHA PEL (United States, 6/2016).

TWA: 0.01 mg/m³, (as Ag) 8 hours.

Appropriate engineering controls : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Environmental exposure controls : Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

Individual protection measures

Hygiene measures : Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Eye/face protection : Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: safety glasses with side-shields.

Skin protection

Hand protection : Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary.

Body protection : Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.

Section 8. Exposure controls/personal protection

- Other skin protection** : Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.
- Respiratory protection** : Based on the hazard and potential for exposure, select a respirator that meets the appropriate standard or certification. Respirators must be used according to a respiratory protection program to ensure proper fitting, training, and other important aspects of use.

Section 9. Physical and chemical properties and safety characteristics

The conditions of measurement of all properties are at standard temperature and pressure unless otherwise indicated.

Appearance

- Physical state** : Solid.
- Color** : Silver. Gray.
- Odor** : Characteristic.
- Odor threshold** : Not available.
- pH** : Not available.
- Melting point/freezing point** : Not available.
- Boiling point, initial boiling point, and boiling range** : 1420 to 1560°C (2588 to 2840°F)
- Flash point** :  Not applicable.
- Evaporation rate** : Not available.
- Flammability** : Not available.
- Lower and upper explosion limit/flammability limit** :  Not applicable.
- Vapor pressure** :  Not available.
- Relative vapor density** :  Not applicable.
- Relative density** : Not available.
- Solubility** : Not available.
- Miscible with water** :  No.
- VOC** : 17.2 g/l
- Partition coefficient: n-octanol/water** :  Not applicable.
- Auto-ignition temperature** :  Not applicable.
- Decomposition temperature** : Not available.
- Viscosity** :  Not applicable.
- Flow time (ISO 2431)** :  Not available.
- Particle characteristics**
- Median particle size** :  Not available.

Section 10. Stability and reactivity

- Reactivity** : No specific test data related to reactivity available for this product or its ingredients.
- Chemical stability** : The product is stable.
- Possibility of hazardous reactions** : Under normal conditions of storage and use, hazardous reactions will not occur.
- Incompatibility with various substances** : Reactive or incompatible with the following materials: oxidizing materials, acids and alkalis.

Continued on next page

Section 10. Stability and reactivity

Hazardous decomposition products : Under normal conditions of storage and use, hazardous decomposition products should not be produced.

Hazardous polymerization : Under normal conditions of storage and use, hazardous polymerization will not occur.

Section 11. Toxicological information

Information on toxicological effects

Acute toxicity

Product/ingredient name	Result	Species	Dose	Exposure
tin	LD50 Oral	Rat	>2000 mg/kg	-
silver	LD Oral	Guinea pig	>5 g/kg	-
	LD Oral	Mouse	>10 g/kg	-
	LD50 Oral	Mouse	100 mg/kg	-
	LD50 Oral	Rat	>2000 mg/kg	-
Proprietary Rosin/Resin	LD50 Dermal	Rabbit	>2.5 g/kg	-
	LD50 Oral	Mouse	>3 g/kg	-
	LD50 Oral	Rat	>4 g/kg	-

Irritation/Corrosion

Not available.

Sensitization

Not available.

Mutagenicity

Not available.

Carcinogenicity

Not available.

Reproductive toxicity

Not available.

Teratogenicity

Not available.

Specific target organ toxicity (single exposure)

Not available.

Specific target organ toxicity (repeated exposure)

Not available.

Aspiration hazard

Not available.

Information on the likely routes of exposure : Inhalation. Ingestion.

Potential acute health effects

Eye contact : No known significant effects or critical hazards.

Inhalation : No known significant effects or critical hazards.

Skin contact : No known significant effects or critical hazards.

Ingestion : No known significant effects or critical hazards.

Symptoms related to the physical, chemical and toxicological characteristics

Eye contact : No specific data.

Inhalation : No specific data.

Continued on next page

Section 11. Toxicological information

Skin contact : No specific data.
Ingestion : No specific data.

Delayed and immediate effects and also chronic effects from short and long term exposure

Short term exposure

Potential immediate effects : Not available.
Potential delayed effects : Not available.

Long term exposure

Potential immediate effects : Not available.
Potential delayed effects : Not available.

Potential chronic health effects

Not available.

General : No known significant effects or critical hazards.
Carcinogenicity : No known significant effects or critical hazards.
Mutagenicity : No known significant effects or critical hazards.
Reproductive toxicity : ☒ No known significant effects or critical hazards.

Numerical measures of toxicity

Acute toxicity estimates

Route	ATE value
☒ Oral	189393.94 mg/kg
Dermal	189393.94 mg/kg

Section 12. Ecological information

Toxicity

Product/ingredient name	Result	Species	Exposure
silver	Acute EC50 1.4 µg/l Marine water	Algae - Chroomonas sp.	4 days
	Acute EC50 0.0092 mg/l	Daphnia	48 hours
	Acute EC50 0.24 µg/l Fresh water	Daphnia - Daphnia magna	48 hours
	Acute LC50 11 µg/l Fresh water	Crustaceans - Ceriodaphnia reticulata	48 hours
	Acute LC50 0.00213 mg/l	Fish	96 hours
	Acute LC50 0.00238 mg/l	Fish	96 hours
	Acute LC50 0.00276 mg/l	Fish	96 hours
	Acute LC50 0.00312 mg/l	Fish	96 hours
	Acute LC50 0.00342 mg/l	Fish	96 hours
	Chronic NOEC 5 mg/l Marine water	Algae - Glenodinium halli	72 hours
Proprietary Rosin/Resin	LC50 60.3 mg/l	Fish	96 hours

Persistence and degradability

Not available.

Bioaccumulative potential

Continued on next page

Section 12. Ecological information

Product/ingredient name	LogP _{ow}	BCF	Potential
silver	-	70	low
Proprietary Rosin/Resin	3.42	-	low

Mobility in soil

Soil/water partition coefficient (K_{oc}) : Not available.

Other adverse effects : No known significant effects or critical hazards.

Section 13. Disposal considerations

Disposal methods : The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

Section 14. Transport information

	DOT Classification	TDG Classification	Mexico Classification	UN	IMDG	IATA
UN number	Not regulated.	Not regulated.	Not regulated.	Not regulated.	Not regulated.	Not regulated.
UN proper shipping name	-	-	-	-	-	-
Transport hazard class(es)	-	-	-	-	-	-
Packing group	-	-	-	-	-	-
Environmental hazards	No.	No.	No.	No.	No.	No.

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Special precautions for user : **Transport within user's premises:** always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

Section 15. Regulatory information

U.S. Federal regulations : TSCA 5(a)2 proposed significant new use rule (SNUR): No products were found.
TSCA 5(a)2 final significant new use rule (SNUR): No products were found.
TSCA 12(b) one-time export notification: No products were found.
TSCA 12(b) annual export notification: No products were found.

United States inventory (TSCA 8b) : All components are listed or exempted.

SARA 302/304

Composition/information on ingredients

No products were found.

SARA 311/312

Classification : Not applicable.

SARA 313

	Product name	CAS number	%
Form R - Reporting requirements	silver	7440-22-4	1-10
Supplier notification	silver	7440-22-4	1-10

SARA 313 notifications must not be detached from the SDS and any copying and redistribution of the SDS shall include copying and redistribution of the notice attached to copies of the SDS subsequently redistributed.

California Prop. 65

 **WARNING:** Cancer - www.P65Warnings.ca.gov.

Canada

Canada inventory : All components are listed or exempted.

International regulations

Inventory list

Australia : Not determined.
China : All components are listed or exempted.
Japan : All components are listed or exempted.
New Zealand : Not determined.
Philippines : Not determined.
Republic of Korea : Not determined.
Taiwan : All components are listed or exempted.

Section 16. Other information

Hazardous Material Information System (U.S.A.)

Health	1
Flammability	0
Physical hazards	0

Procedure used to derive the classification

Not classified.

History

Date of issue/Date of revision : 3/10/2023
Date of previous issue : 1/23/2022
Version : 1.01

Continued on next page

Section 16. Other information

Regulatory Affairs Department
enthone.msds@macdermidenthone.com

Key to abbreviations

: ATE = Acute Toxicity Estimate
BCF = Bioconcentration Factor
GHS = Globally Harmonized System of Classification and Labelling of Chemicals
IATA = International Air Transport Association
IBC = Intermediate Bulk Container
IMDG = International Maritime Dangerous Goods
LogPow = logarithm of the octanol/water partition coefficient
MARPOL = International Convention for the Prevention of Pollution From Ships, 1973
as modified by the Protocol of 1978. ("Marpol" = marine pollution)
N/A = Not available
SGG = Segregation Group
UN = United Nations

References

: Not available.

Indicates information that has changed from previously issued version.

Notice to reader

To the best of our knowledge, the information contained herein is accurate. However, neither the above-named supplier, nor any of its subsidiaries, assumes any liability whatsoever for the accuracy or completeness of the information contained herein.

Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

4.9.04b4933

Kester SDS GHS Americas

SAFETY DATA SHEET LITHIUM IRON PHOSPHATE (LiFePO4)

This Data Sheet is not intended to be a comprehensive exposition of the properties of Lithium Iron Phosphate batteries. No guarantee or warranty of any kind, express or implied, is made with respect to the information contained herein.

1. COMPANY AND PRODUCT IDENTIFICATION**Product Name:** LiFePO4 Rechargeable Battery**Chemical System:** LiFePO4**Shipping:** Packages <66lb Ground Only,
Packages >66lb Freight Only**Distributed By:** EXPION360

2025 SW Deerhound Ave.

Redmond, OR 97756

Phone Number: 541-797-6714**Common Name:** Lithium Iron Phosphate Battery**(LiFePO4) Product Use:** Electric Storage Battery**Proper Shipping Name:** UN3480, Lithium Battery, Class
9, Packing Group 1.**Email:** support@expion360.com**Fax Number:** 541-797-6715**Emergency Number:** 800-424-9300 Chemtrec 990142**2. HAZARDS IDENTIFICATION****Emergency Overview:**

Lithium Iron Phosphate Batteries are not dangerous with normal use. The materials within the battery may only represent a hazard if the structural integrity of the battery is compromised or the battery is subjected to extreme misuse. Do not expose the batteries to fire or open flame. Do not mix batteries of varying sizes, chemistries, or types. Do not short circuit, puncture, incinerate, crush, over-charge, over-discharge, or expose the batteries to temperatures above or below the declared limit. Damage to the batteries may result in the risk of fire or explosion, which could release dangerous hydrogen fluoride gas and exposure to the ingredients contained within.

Potential Health Effects:

Acute (Short Term): see Section 8 for Exposure Controls and Personal Protection. If the battery is disassembled or ruptured, the electrolyte in the cells is corrosive and can cause burns to the skin and eyes.

Potential Health Effects: Inhalation

If the battery remains sealed, there is no expected exposure. If the battery has been ruptured, vapor or mists can cause respiratory issues.

Potential Health Effects: Ingestion

If the battery remains sealed, there is no expected exposure to any chemical or hazard to be ingested. If the battery has been ruptured, mists and vapors can enter the mouth and may cause respiratory irritation/ chemical burns of the mouth and gastrointestinal tract irritation.

Potential Health Effects: Skin

Contact with a sealed battery by skin will cause no harm from any internal chemicals. Skin contact with positive and negative terminals of high voltages may cause burns to the skin. Skin contact with a ruptured battery can cause skin irritation.

Potential Health Effects: Eyes

Eye contact with the contents of a ruptured battery can cause severe irritation to the eye.

HMIS Ratings: Health: 0 Fire: 0 HMIS Reactivity: 0

Hazard Scale: 0 = Minimal 1 = Slight 2 = Moderate 3 = Serious 4 = Severe * = Chronic hazard

3. COMPOSITION / INFORMATION ON INGREDIENTS

IMPORTANT NOTE: Safety information is given for exposure to the product as sold. The intended use of the product should not result in exposure to the chemical substance. The battery cell should not be opened or exposed to heat as exposure to the ingredients contained within could be harmful under some circumstances.

Concentration %	Components	CAS No.
22.4-24.8	Ferrous Phosphate Lithium	15365-14-7
0.7-0.9	Polyvinylidene fluoride resin	24937-79-9
0.2	Cellulose, carboxymethyl ether	9000-11-7
0.3-0.4	Styrene-butadiene rubber 1500	9003-55-8
12-13.2	Carbon black	1333-86-4
1.6-2.4	Polypropylene	9003-07-0
1.2-1.4	Phosphate (1-), hexafluoro-, lithium	21324-40-3
2.4-2.9	Carbonic acid, dimethyl ester	616-38-6
4.0-5.6	Carbonic acid, ethyl methyl ester	623-53-0
2.0-2.4	1, 3-Dioxolan-2-one	96-49-1
2.0	Polyvinyl chloride resin	9002-86-2
7.2-8.0	Copper	7440-50-8
3.2-3.6	Aluminum	7429-90-5
16-19.2	Nickel	7440-02-0
1.6-4.7	Polybutadiene, low molecular (1, 2-lpb)	25038-44-2
4.5-9.3	Styrene-maleic anhydride copolymer	9011-13-6

*Concentration % is a trade secret and exact measurements cannot be supplied.

4. FIRST-AID MEASURES

Skin Contact:

Contact with internal contents may cause burns. If skin contact with internal contents occurs, remove affected articles of clothing. Wash the affected area with lukewarm water for at least 30 minutes. If irritation or pain persists, seek medical attention. Decontaminate affected articles of clothing before reuse or discarding.

Eye Contact:

Contact with internal contents may cause burns. If eye contact with internal contents occurs, wash out the affected eye with gently flowing lukewarm water while holding the eyelids open for at least 30 minutes. Rinse with neutral saline solution if possible. Use caution not to rinse contaminated water into the unaffected eye, nose, mouth, or face. Seek medical attention.

Inhalation:

If internal contents are inhaled, move the victim to fresh air and remove the source of contamination from the area. Seek medical advice.

Ingestion:

If ingestion of internal contents occurs, rinse your mouth thoroughly with water. DO NOT INDUCE VOMITING. If vomiting occurs naturally, have the victim lean forward to reduce the risk of aspiration and continue to rinse the mouth with water. Seek medical attention immediately.

Caution:

In all cases, evacuate the contaminated area. If irritation persists, seek medical assistance at once.

5. Fire-Fighting Measures

Suitable Extinguishing Media:

Lithium-ion batteries (including LiFePO_4 , etc...) are a rechargeable type of lithium battery. Because these batteries do not contain a high concentration of lithium metal, you can use water, a class ABC fire extinguisher, or a CO_2 extinguisher. Do not use a class D extinguisher with Lithium-ion batteries. Water, carbon dioxide, dry chemical powder, and foam are the most effective means to extinguish a battery fire.

Fire Fighting Procedure:

Wear fully protective gear, including self-contained positive pressure breathing apparatus, goggles, fireproofing jacket, and gloves. Caution is advised during water application because burning particles may be ejected from the fire.

Unusual Fire and Explosion Hazards:

Exposing battery cells to excessive heat, fire, or over-voltage conditions may cause a leak, fire, hazardous vapors, and hazardous decomposition products. Damaged or opened cells or batteries can result in rapid heating and the release of flammable vapors and potentially dangerous gases that may be heavier than air and could travel along the ground or be moved by ventilation to an ignition source.

6. Accidental Release Measures

Personal Precautions:

Hazardous material contained within the battery's cells will only be expelled if the battery is damaged or abused. If an accidental release occurs, personnel in the immediate vicinity should ensure containment measures and evacuation procedures are performed rapidly before any cleanup. All non-required personnel for containment and clean-up should observe the evacuation procedures.

Evacuation Procedures:

If an accidental release occurs, evacuate the area except for required containment and clean-up personnel. Maintain a minimum clearance of 25 meters (75 feet) in all directions. Stay upwind of the release, keep out of low areas, and ventilate closed areas before re-entering.

Environmental Precautions:

Prevent released material from contaminating soil or entering sewers or waterways by capping drains or placing up barriers.

Containment Procedures:

Stop the release if it is safe to do so. Contain any spilled liquid with dry sand, earth, or vermiculite. Move the damaged object to an isolated area or containment chamber or cover it with a fireproof containment blanket if it is safe to do so. Clean up spills immediately.

Clean Up Procedures:

Wear adequate personal protective equipment as indicated in Section 8. Absorb spilled liquid material with an inert absorbent (dry sand, earth, or vermiculite) material. Collect all debris and contaminated absorbents into an acceptable waste container and dispose of them according to the directions in Section 13. Scrub the spill area with detergent and water; collect all contaminated wash water for proper disposal.

7. Handling and Storage

Handling Precautions:

Do not expose the battery or cell to extreme temperatures or fire. Do not disassemble, crush, or puncture the battery. Do not overcharge or over-discharge the battery. Do not mix batteries of varying types or sizes. Do not connect (short circuit) positive and negative terminals or place the batteries on conductive metal.

Safe Storage Recommendations:

Insulate positive and negative terminals when not in use to avoid short circuits. Ensure sufficient clearance between batteries and other surfaces. Store in a dry, cool (70°F +/-5°F) and well-ventilated area. Elevated temperatures can result in reduced battery life and venting of flammable liquid and gases. Keep out of reach of children.

8. Exposure Controls/Personal Protection

Personal Protection:

- Respiratory Protection: Not necessary under normal use. In case of battery rupture, provide as much ventilation as possible. Avoid confined areas where cells are venting. Use a self-contained full-face respirator mask.
- Skin Protection: Not necessary under normal use. Wear a rubber apron and gloves if handling a ruptured or leaking cell.
- Eye Protection: Not necessary under normal use. Wear safety goggles if handling ruptured or leaking cells.
- Engineering Controls: Use local exhaust ventilation and keep out of confined areas with ruptured or leaking batteries.

9. Physical and Chemical Properties

Appearance: Cell Battery	Physical State: Solid
Color: Not Applicable	pH: Not Applicable
Odor Type: Odorless	Odor Threshold: Not Applicable
Freezing Point: Not Applicable	Melting Point: Not Applicable
Boiling Point: Not Applicable	Boiling Range: Not Applicable
Flash Point and Method (CO): Not Applicable	Evaporative Rate: Not Applicable
Flammability: Not Applicable	Flammability/Explosive Limits (%): Not Applicable
Decomposition Temperature:	Viscosity: Not Applicable
Relative Density: Not Applicable	Auto Ignition Temperature (CO): Not Applicable
Solubility in Water: Not Applicable	Vapor Pressure: Not Applicable
Partition Coefficient: Not Applicable	Vapor Density: Not Applicable

10. Stability and Reactivity

Reactivity: Not Available

Chemical Stability: The product is stable under the conditions described in Section 7. Other: Possibility of producing hazardous gas in reaction with water.

Conditions to Avoid: Do not expose to high temperatures, incinerate, deform, mutilate, crush, pierce, short circuit, disassemble, or humid conditions (for long periods).

Incompatible Materials: Oxidizing agents, alkalis, and water.

Hazardous Decomposition Products: Toxic fumes and possible peroxides.

11. Toxicological Information

Likely routes of exposure: Normal use will result in unlikely routes to any exposure.

Delayed, Immediate, or Chronic effects from exposure: Normal use will result in no known effects from exposure.

Other Toxicity and Effect Information:

Irritation: The risk of irritation only occurs if battery cells are mechanically, thermally, or electrically damaged and the enclosure is compromised. If this occurs, irritation to the skin, eyes, and respiratory tract may occur.

Sensitization: The nervous system and organs may be sensitized by exposure to a compromised battery cell enclosure.

Carcinogenicity: Normal use will not result in exposure to substances that are considered human carcinogens by IARC (International Agency for Research on Cancer), ACGIH (American Conference of Governmental Industrial Hygienists), OSHA or NTP (National Toxicology Program).

The information on the internal cell materials is as follows:

Lithium Iron Phosphate (LiFePO₄)

Acute toxicity: No applicable data. Local effects: Unknown. Sensitization: The nervous system of respiratory organs may become sensitive. Chronic toxicity/Long-term toxicity: No applicable data. Skin causticity: Although it is very rare, a rash of the skin and allergic erythema may result.

Aluminum

Local effects: Aluminum itself has no toxicity. When it goes into a wound, dermatitis may be caused. Chronic toxicity/Long term toxicity: By the long-term inhalation of coarse particulate or fume, it is possible to cause lung damage (aluminum lungs).

Graphite

Acute toxicity: Unknown.

Local effects: When it goes into the eyes, it stimulates the eyes; conjunctivitis, thickening of corneal epithelium, or edematous inflammation of the palpebra may be caused.

Chronic toxicity/Long-term toxicity: Since the long-term inhalation of high levels of graphite coarse particulate may become a cause of a lung disease or a tracheal disease.

Carcinogenicity: Graphite is not recognized as a cause of cancer by research organizations and natural toxic substance research organizations of cancer.

Copper

Acute toxicity: 60-100mg sized coarse particulate causes a gastrointestinal disturbance with nausea and inflammation.

TDLo, hypodermic - Rabbit 375mg/kg Local effects: Coarse particulate stimulates the nose and throat. Eyes will become red and painful if contact is made.

Sensitization: Sensitization of the skin may be caused by long-term or repetitive contact.

Reproductive effects: TDLo, oral - Rat 152mg/kg

12. Ecological Information (non-mandatory)

Mammalian effects: None known at this time.

Eco-toxicity: None known at this time.

Bioaccumulation potential: Slowly Bio-degradable.

Environmental Fate: None known environmental hazards at this time.

13. Disposal Considerations

Waste Disposal Method:

Recycling is encouraged. DO NOT dump into sewage or water bodies. Dispose of in accordance with local, state, and federal laws and regulations. DO NOT Incinerate.

Special Precautions:

Discharge batteries fully and cap terminals before disposal. Handle according to Section 7 and Section 8 to minimize exposure.

Regional Regulations:

USA: Dispose of in accordance with local, state, and federal laws and

regulations. Other: Dispose of in accordance with local, state, and federal laws and regulations.

14. Transportation Information (non-mandatory)

IATA Classification: (International Air Transportation Association)

UN Number: UN3480

UN Proper Shipping Name: LITHIUM ION BATTERIES

Transport Hazard Class: Class 9

Packing Group Number: Packing Group IA

Notes and Exceptions: Packaging, markings, and documentation requirements are defined in the International Air Transport Association (IATA) Dangerous Goods Regulations (DGR) Packing Instructions 965.

IMDG Classification: {International Maritime Dangerous Goods}

UN Number: UN3480

UN Proper Shipping Name: LITHIUM ION BATTERIES

Transport Hazard Class: Class 9

Packing Group Number: Packing Group IA

Notes and Exceptions: Packaging, markings, and documentation requirements are defined in the IMDG code Packing Instructions P903.

U.S. HMR Classification: {United States Hazardous Materials

Regulations} UN Number: UN3480

UN Proper Shipping Name: LITHIUM BATTERY

Transport Hazard Class: Class 9

Packing Group Number: Packing Group IA

Notes and Exceptions: Packaging, markings, and documentation requirements are defined in Title 49 of the Code of Federal Regulations (CFR), Section 173.185. of the U.S. HMR.

15. Regulatory Information (non-mandatory) USA

OSHA HCS: This SDS complies with requirements of the Hazard Communication Standard {HCS} 29 CFR 1910.1200{g} and Appendix D

EPA TSCA Status: All ingredients in the product are listed on the TSCA inventory. EPA SARA Title III:

Sec. 302/304: None Sec. 311/312: None Sec. 313: None

EPA CERCLA RQ: None

California Prop 65: Warning: This product contains chemicals known to the State of California to cause cancer and birth defects or other reproductive harm. For more information: www.P65Warnings.ca.gov

16. Other Information

This information has been compiled from sources considered to be dependable and is, to the best of our knowledge and belief, accurate and reliable as of the date compiled. However, no representation, warranty (either expressed or implied), or guarantee is made to the accuracy, reliability, or completeness of the information contained herein.



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Home > Z. LiMnNi Batteries

Category : Z. LiMnNi Batteries

Knowledge: There are three types of Li-Ion cells on current market based on chemistry of cathode materials. Please see the table below to see advantage and disadvantage of each type cell. Among of them, LiCoO₂ series Li-Ion cell has the highest energy density. LiFePO₄ is the most safe but lower energy density and LiMnO₄ series cell is safer and higher enegidensity, which may be more suitable for Electric Vehicle

Chemistry	Voltage	Energy Density	Working Temp.	Cycle Life	Safety	Environmental	Cost based on cycle life x wh of SLA
LiFePO ₄	3.2V	>120 wh/kg	-0-60 °C	>2000	Safe	Good	0.15-0.25 lower than SLA
Lead acid	2.0V	> 35wh/kg	-20 - 40°C	>200	Safe	Not good	1
NiCd	1.2V	> 40wh/kg	-20 - 50 °C	>1000	Safe	Bad	0.7
NiMH	1.2V	>80 wh/kg	-20 - 50 °C	>500	Safe	Good	1.2-1.4
LiMn _x Ni _y Co ₂ O ₂	3.7V	>160 wh/kg	-20 - 40 °C	>500	Unsafe without PCB or PCM, better than LiCo	OK	1.5-2.0
LiCoO ₂	3.7V	>200 wh/kg	-20 - 60 °C	> 500	UnSafe without PCB or PCM	OK	1.5-2.0

[3.7V LiMnNi Single Cells](#)

[11.1V LiMnNi Battery Packs](#)

[14.8V LiMnNi Battery Packs](#)



[18.5V LiMnNi Battery Packs](#)

[25.9V LiMnNi Battery Packs](#)

[37V - 88.8V LiMnNi Battery Packs](#)



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Emergency Light Batteries
Energy Storage / Solar Power
EV/Golf Car/PowerSport
Flashlight / Batteries
Fluke Replacement Batteries
Home Security Batteries
LED Lights / Display
Mobile Electronics
PowerTool Batteries
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DC Motor/Control Circuit

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Category : Li-Ion/Polymer Single Cells

Knowledge

There are three types of Li-Ion cells on current market based on chemistry of cathode materials. Please see the table below to see advantage and disadvantage of each type cell. Among of them, LiCoO₂ series Li-Ion cell has the highest energy density and is most popular cell using in the market. LiFePO₄ and LiMnNiO₄ cell still are in developing

Chemistry	Voltage	Energy Density	Working Temp.	Cycle Life	Safety	Environmental	Cost based on cycle life x wh of SLA
LiFePO ₄	3.2V	>120 wh/kg	-0-60 °C	>2000	Safe	Good	0.15-0.25 lower than SLA
Lead acid	2.0V	> 35wh/kg	-20 - 40°C	>200	Safe	Not good	1
NiCd	1.2V	> 40wh/kg	-20 - 50 °C	>1000	Safe	Bad	0.7
NiMH	1.2V	>80 wh/kg	-20 - 50 °C	>500	Safe	Good	1.2-1.4
LiMn _x Ni _y Co ₂ O ₂	3.7V	>160 wh/kg	-20 - 40 °C	>500	Unsafe without PCB or PCM, better than LiCo	OK	1.5-2.0
LiCoO ₂	3.7V	>200 wh/kg	-20 - 60 °C	> 500	UnSafe without PCB or PCM	OK	1.5-2.0

Warnings

- Li-Ion battery is dangerous. Must read [safety instructions](#) before buying / using Li-ion / Polymer battery and packs
- Never use Li-Ion battery pack without [PCB](#) (protection circuit board).
- Please choose our [smart chargers](#) to recharge a Li-Ion battery pack

[3.6V / 3.7V Li-Ion Cylindrical Cell Series](#)



18650(4/3AF), 18500,
14500(AA), 14430(4/5AA)
10440(AAA), 14650 (7/5AA),
17500(A), 20700, 21700, 26650
(Long C), 26650M, 38120P(M),
40160S(Long M), RCR123A,

[3.7V Polymer Li-Ion Single Cells](#)



1-2C rate, High Power, UL listed,
& Others size Polymer Li-Ion
Batteries

[3.6V Li-Ion Rechargeable Button Cells & Charger](#)



2032, 2450 Button Cells with tabs /
without tabs & Smart Charger

[9V Li-Ion Rechargeable Cells & Smart Charger](#)



FAQ

Q: Your data sheet and warning pages recommend charge the polymer battery between 0 degC and 40 degC. What sort of effects will we expect to see if we charge the battery in ambient conditions near or above 40 degC with little to no air movement? Will this affect performance only or are we at a high risk of swelling or possible venting?

A: The spec is from our supplier. This means under these temperature the cell is guaranteed to work or charge w/o problem. If you charge at higher temperature, it could cause the cell to swollen and damage the cell.

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Energy Storage / Solar Power
EV/Golf Car/PowerSport
Flashlight / Batteries
Fluke Replacement Batteries
Home Security Batteries
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Mobile Electronics
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Category : LTO Batteries

The **lithium-titanate battery** (Li4Ti5O12, referred to as LTO in the battery industry) is a type of rechargeable battery based on advanced nano-technology, which has the following advantages than other lithium batteries.

Advantages:

- Li-Titanate batteries have a wider operating temperature range (**Charge: 0-45°C; Discharge: -30 to 70°C**) and a recharge efficiency exceeding 98%, compared to other carbon based batteries.
- Li-Titanate batteries as long cycle life: **> 3000 cycles**
- Li-Titanate batteries are high security, high stability.
- Li-Titanate batteries are green & eco-friendly.

Disadvantage:

The disadvantage is that lithium-titanate batteries have a lower inherent voltage (**2.4V/cell**), which leads to a lower energy density than conventional lithium-ion battery technologies. But the energy density of LTO - based batteries is still higher than lead acid and NiCad batteries.

Applications: Ideal for High Rate and High Cycle Life Applications

Because of the benefits of lithium titanate in terms of high security, high stability, long life and green features, lithium titanate batteries can be widely used in military, aerospace, electric vehicles and charging stations, tourist coaches, yachts, wind and solar energy storage power, traffic signals, solar hybrid street lighting, UPS power supply, home storage, coal, disaster relief emergency, weather radar, electricity, smart grid, communication base stations, hospitals, finance, telecommunications as well as system critical backup power systems.

Displaying products 1 - 4 of 4 results

Show: 30 Sort: Default



LTO 1020 Rechargeable Cell:
2.4V 50 mAh, 0.12Wh, 2A rated

Your Price: \$8.63

Product ID: 7455



LTO 1450 Rechargeable Cell:
2.4V 500 mAh, 1.2Wh, 15A rated

Your Price: \$11.44

Product ID: 7470



LTO 1865 Rechargeable Cell:
2.4V 1300 mAh, 39A rated
3.12Wh - UN38.3 Passed

Your Price: \$14.89

Product ID: 7472



LTO M55380S Rechargeable
Cell: 2.4V 45Ah, 108Wh, 135A
rated

Your Price: \$287.44

Product ID: 9445



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Lithium-ion Battery Safety

Lithium-ion batteries are one type of rechargeable battery technology (other examples include sodium ion and solid state) that supplies power to many devices we use daily. In recent years, there has been a significant increase in the manufacturing and industrial use of these batteries due to their superior energy storage characteristics. This increased use of lithium-ion batteries in workplaces requires an increased understanding of the health and safety hazards associated with these devices.

The hazards and controls described below are important in facilities that manufacture lithium-ion batteries, items that include installation of lithium-ion batteries, energy storage facilities, and facilities that recycle lithium-ion batteries.

Lithium-ion Batteries

A lithium-ion battery contains one or more lithium cells that are electrically connected. Like all batteries, lithium battery cells contain a positive electrode, a negative electrode, a separator, and an electrolyte solution. Atoms or molecules with a net electric charge (i.e., ions) are transferred from a positive electrode to a negative electrode through an electrolyte solution. Lithium cells store and release power by converting chemical potential energy into electrical energy using lithium ions or lithium metal. Electrolyte solutions allow ions to flow freely between the electrodes. There are several types of lithium cells, including cylindrical cells, prismatic pouch cells, and prismatic metal can cells.

Lithium-ion batteries use lithium in ionic form instead of in solid metallic form and are usually rechargeable, often without needing to remove the battery from the device. They power devices such as mobile telephones, laptop computers, tablets, cameras, power tools, electric vehicles, and machinery, and are also used in large Energy Storage Systems (ESS).

Potential Hazards

Lithium-ion batteries may present several health and safety hazards during manufacturing, use, emergency response, disposal, and recycling. These hazards can be associated with the chemicals used in the manufacture of battery cells, stored electrical energy, and hazards created during thermal runaway, (see below) which can include fire, explosions, and chemical byproducts.



Chemical Hazards

Lithium-ion batteries contain various components that present different chemical hazards to workers, such as flammability, toxicity, corrosivity, and reactivity hazards. These chemicals may enter the workplace as raw materials or recycled materials. As processes change, any new chemicals must be thoroughly assessed for potential safety and health impacts to the workplace and workers.

A lithium-ion battery cathode is made of a lithium metal oxide material. The choice of cathode material depends on the desired characteristic of the battery. These materials can include lithium cobalt oxide (LiCoO_2), lithium manganese oxide (LiMn_2O_4), lithium nickel manganese cobalt oxide (LiNiMnCoO_2), lithium nickel cobalt aluminum oxide (LiNiCoAlO_2), or lithium iron phosphate (LiFePO_4).

Common materials for a lithium-ion battery anode include carbon-based materials such as graphene, nanofibers, carbon nanotubes, graphite, and titanium-based materials such as lithium titanate and titanium dioxide.

Lithium-ion batteries contain electrolytes that are a combination of solvents with an electrolytic salt. Lithium hexafluorophosphate, the most common salt used in lithium-ion cells, can react with water to form hydrogen fluoride (HF). The most common solvents used in lithium-ion batteries include ethylene carbonate (EC), propylene carbonate (PC), dimethyl carbonate (DMC), ethyl methyl carbonate (EMC), and diethyl carbonate (DEC). Some of these electrolytes are flammable liquids and requirements within OSHA's [Process Safety Management](#) standard may apply to quantities exceeding 10,000 lb.

Many of the chemicals used in lithium-ion battery manufacturing have been introduced relatively recently. Consequently, there may be limited toxicological information and few established OSHA permissible exposure limits (PELs). Additionally, because some of OSHA's PELs may be outdated and inadequate for protecting worker health, employers should consider alternative occupational exposure limits (OELs) developed by technical, professional, industrial and/or government organizations to ensure worker protection. For chemicals without occupational exposure limits, Occupational Exposure Banding and Control Banding can be used with the [Hierarchy of Controls](#) to manage risks and prevent exposure to hazardous chemicals. See the [NIOSH Control Banding](#) webpage and the OSHA Permissible Exposure Limits – Annotated Tables for additional information and explanation.

Additionally, when a lithium-ion battery or cell does not meet exemptions under OSHA's [Hazard Communication Standard \(HCS\)](#) as an "article," the manufacturer or importer is required to classify the chemical hazards and provide the hazard information to downstream users. For additional information, see OSHA's Letters of Interpretation regarding the [Coverage of lithium-ion batteries under the Hazard Communications Standard \(6/23/2021\)](#) and [Applicability of the HCS to Lithium-ion Batteries \(12/1/2022\)](#).

Safety Hazards

In addition to electrical hazards, lithium-ion batteries can also present hazards resulting from thermal runaway. Because lithium-ion batteries combine a flammable electrolyte with a significant amount of stored energy, thermal runaway reactions are possible. Thermal runaway is a chain reaction where the heat released from the failure of one cell damages nearby cells. This can be initiated by internal short circuiting due to defects during manufacturing, mechanical damage to the battery, exposure to excessive heat or cold, and improper charging.

Thermal runaway can be identified by several indicators including a rise in battery temperature, venting of gas, vapor, or smoke from the battery, or the presence of fire. Fires caused by thermal runaway can produce additional chemical hazards that may include hydrogen fluoride (HF), hydrogen chloride (HCl), hydrogen cyanide (HCN), phosphoryl fluoride (POF₃), carbon monoxide (CO), carbon dioxide (CO₂), black carbon, and other potentially hazardous chemicals and particulates.

For additional information see OSHA's Safety and Health Information Bulletin on [Preventing Fire and/or Explosion Injury from Small and Wearable Lithium Battery Powered Devices](#).

Safety by Design

Whether manufacturing or using lithium-ion batteries, anticipating and designing out workplace hazards early in a process adoption or a process change is one of the best ways to prevent injuries and illnesses. Hazard controls must be addressed in the initial design and construction phases, as well as implemented through changes to management and manufacturing processes as production methods and energy storage technologies evolve. Safety by design includes the proactive substitution and adoption of less hazardous technologies. See the NIOSH webpage, [Prevention through Design](#), for additional information.

Safety and Health Management System

Establishing a safety and health management system (SHMS) (i.e., safety program) is an effective way of protecting workers from potential hazards associated with lithium-ion

batteries. A mature and effective SHMS can prevent workplace injuries and illnesses by using proactive approaches to find and fix workplace hazards before they cause injury and illness to workers. To be effective, the SHMS must have meaningful worker engagement and participation. In many workplaces, worker participation may include a safety committee that can bring workers and management together to identify and find solutions, and to promote safety and health. See the OSHA webpage [Recommended Practices for Safety and Health Programs](#) for additional information on implementing a successful SHMS or program.

Hazard Controls

Lithium-ion battery hazard controls should be implemented according to the [Hierarchy of Controls](#). Controlling hazards at the source is the most effective method to eliminate or reduce hazards. . OSHA's [Transitioning to Safer Chemicals Toolkit](#) is a step-by-step resource with information, methods, tools, and guidance for employers and workers to proactively reduce or eliminate chemical hazards at the source through informed substitution.

Additionally, hazard controls that can be implemented in workplaces that manufacture or use lithium-ion batteries include:

- Ventilation, including local exhaust ventilation (LEV) and enclosures
- Process automation and isolation of hazardous materials
- Storage of lithium-ion batteries and devices in dry, cool locations
- Following National Fire Protection Association (NFPA) guidance for the installation of [Energy Storage Systems](#)
- Following manufacturer's instructions for storage, use, charging, and maintenance of lithium-ion batteries
- Limiting the quantity of stored lithium-ion batteries
- Following the manufacturer's guidance on how to extinguish small battery fires
- Continuous monitoring for flammable and toxic gases in large storage locations
- Using shipping guidance provided by the [US Department of Transportation](#) and [International Air Transport Association](#) (IATA) and following proper shipping and packaging requirements provided in [49 CFR 173.185](#)

- Disposing of lithium-ion batteries and devices containing these batteries, at designated recycling facilities and not placing them in municipal trash or recycling bins
- Not mixing battery types (e.g. lithium-ion, alkaline, lead acid) in recycling facilities
- Providing safety showers and eyewash stations in locations where exposure to electrolytes may occur
- Developing and implementing an emergency response plan, including emergency procedures, and creating training for response personnel that addresses possible physical and chemical hazards, including hazardous decomposition products (e.g., hydrogen fluoride)
- Conducting a hazard assessment and using proper personal protective equipment (PPE), when appropriate

Training

Education and training are important tools for informing workers and managers about workplace hazards and controls. Workers must be trained in a language and at a literacy level they understand. Appropriate training may include, but is not limited to, [Hazard Communication](#) (29 CFR 1910.1200) training, [Hazardous Waste Operations and Emergency Response \(HAZWOPER\)](#) (29 CFR 1910.120) training, and [Process Safety Management](#) (29 CFR 1910.119) training . OSHA's publication [Training Requirements in OSHA Standards](#) provides a general overview of the training requirements in specific OSHA standards.

In workplaces with lithium-ion batteries, it is important that employers ensure that an emergency action plan (EAP) includes lithium-related incident response procedures based on the manufacturer's instructions and NFPA guidance for responding to battery failures, including fires and/or explosions caused by thermal runaway, and that workers are trained on these procedures.

For workplaces involved in the manufacturing, repair, use, and recycling of lithium-ion batteries, it is important for employers to ensure that exposed workers receive appropriate information about the hazards associated with lithium-ion batteries and that workers receive training on the physical and health hazards associated them.

OSHA Standards

While there is not a specific OSHA standard for lithium-ion batteries, many of the OSHA general industry standards may apply, as well as the General Duty Clause (Section 5(a)(1) of the Occupational Safety and Health Act of 1970). These include, but are not limited to the following standards:

- 1910 Subpart L - Fire Protection
- 1910 Subpart S - Electrical
- 1910.39 Fire Prevention Plans
- 1910.119 Process Safety Management of Highly Hazardous Chemicals
- 1910.120 Hazardous Waste Operation and Emergency Response
- 1910.132 Personal Protective Equipment
- 1910.134 Respiratory Protection
- 1910.147 The Control of Hazardous Energy (Lockout/Tagout)
- 1910.151 Medical Services and First Aid
- 1910.178 Powered Industrial Trucks
- 1910.1000 Air Contaminants
- 1910.1020 Access to employee exposure and medical records
- 1910.1200 Hazard Communication

These standards are also representative of the types of protections that apply to the manufacture and use of other energy storage technology, whether in use now or under development.

Consensus/Industry Standards and Programs

- National Fire Protection Association, NFPA 855 Standard for the Installation of Stationary Energy Storage Systems
- International Electrotechnical Commission, IEC 62281 Safety of Primary and Secondary Lithium Cells and Batteries During Transport
- Underwriters Laboratories, UL 2054 Standard for Household and Commercial Batteries
- Underwriters Laboratories, UL 9540 Standard for Energy Storage Systems and Equipment
- The American Society of Safety Professionals, ANSI/ASSP Z10.0 Occupational Health and Safety Management Systems
- International Organization for Standardization, ISO 45001 Occupational Health and Safety Management Systems

How to Contact OSHA

OSHA's mission is to assure America's workers have safe and healthful working conditions free from unlawful retaliation. For more information, visit www.osha.gov or call OSHA at 1-800-321-OSHA (6742), TTY 1-877-889-5627.

This is one in a series of informational fact sheets highlighting OSHA programs, policies or standards. It does not impose any new compliance requirements. For a comprehensive list of compliance requirements of OSHA standards or regulations, refer to Title 29 of the Code of Federal Regulations. This information will be made available to sensory-impaired individuals upon request. The voice phone is (202) 693-1999; teletypewriter (TTY) number: (877) 889-5627.





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Home > PCB/BMS/CMB/UPS

Category : PCB/BMS/CMB/UPS

PCM or PCB (protection circuit module or board) is the "heart" of Lithium battery pack. It will protect Lithium battery pack from overcharging, over-discharging and over-drain, therefore it is the must to have to avoid Lithium battery pack from explosion, fire and damage.

For low voltage lithium battery packs (<20 cells), you shall choose PCM with equilibrium function to keep each cell in balance and good service life. For high voltage Lithium battery pack (>20 cells), you shall consider advanced BMS (battery manage system) to monitor each cell's performance to ensure battery's safer operation.

A battery management system (BMS) is any electronic system that manages a rechargeable battery (cell or battery pack), such as by monitoring its state, calculating secondary data, reporting that data, protecting the battery, controlling its environment, and / or balancing it. A BMS may monitor the state of the battery as represented by various items, such as:

- Voltage: total voltage, or voltages of individual cells
- Temperature: average temperature, or temperatures of individual cells
- State of charge (SOC) or depth of discharge (DOD): to indicate the charge level of the battery
- State of health (SOH), a variously-defined measurement of the overall condition of the battery
- Current: current in or out of the battery
- New Feature: [Bluetooth Technology for Lithium-ion/LiFePO4 batteries is available for battery protection circuit board](#)

AA Portable power corp. has our own factory to design and manufacture high quality PCM and BMS for all types Lithium battery packs. We test each PCM before shipping and promise PCM is functional and safe. If you have any question in using PCM, please feel free to contact us at tech@batteryspace.com

[PCM for LiCo / LiMnNi / LiNiCoMn Battery Packs](#)



From 3.7V-88.8V

[PCM for LiFePO4 Battery Packs](#)



From 3.2V-76.8V, & BMS

[PCM with Bluetooth/SMBus](#)



For LiCo/LiFePO4/LiMnNi/LiNiCoMn/

[BMS \(Battery Management System\) for Li-Ion/LiFePO4 Pack](#)



[Custom PCB/PCM for Li-Ion/LiFePO4 battery pack](#)



[PCB for NiMH Battery Pack](#)



[CMB \(Charge Management Board\) for Li-Co / Li-Mn-Ni / LiFePO4 Packs](#)



[DC-DC regulated Converters](#)



[UPS Battery Back Up Board for LiFePO4 / SLA Battery Pack](#)



News

[What to consider when designing Lithium Batteries?](#)

[Bluetooth Technology for Lithium-ion/LiFePO4 Batteries](#)

[UN 38.3 Safety Test](#)

We own a national standard [UN38.3 testing lab](#). We can provide [UN38.3 test service](#) to our customer.

If you want to set up your own lab, we also provide [UN38.3 & IEC62133 equipments](#) and onsite training. Email us for the quote and lead time.

[Lithium Battery Shipping Regulation](#)

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Lithium-Ion Battery Safety



Lithium-ion batteries supply power to many kinds of devices including smart phones, laptops, e-scooters and e-bikes, e-cigarettes, smoke alarms, toys, and even cars. If not used correctly, or if damaged, these batteries can catch on fire or explode.

The Problem

- These batteries store a large amount of energy in a small amount of space.
- Sometimes batteries are not used the right way; batteries not designed for a specific use can be dangerous.
- Like any product, a small number of these batteries are defective. They can overheat, catch fire, or explode.

Safety Tips

- Purchase and use devices that are listed by a nationally recognized testing laboratory.
- Always follow the manufacturer's instructions.
- Only use the battery that is designed for the device.
- Only use the charging cord that came with the device.
- Do not charge a device under your pillow, on your bed, or on a couch.
- Do not keep charging the device or device battery after it is fully charged.
- Keep batteries at room temperature when possible. Do not charge them at temperatures below 32°F (0°C) or above 105°F (40°C).
- Store batteries away from anything that can catch fire.

Signs of a Problem

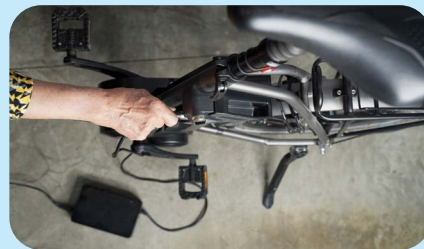
Stop using the battery if you notice these problems: odor, change in color, too much heat, change in shape, leaking, or odd noises. If it is safe to do so, move the device away from anything that can catch fire. Call **9-1-1**.

Recycle Batteries Responsibly

Don't throw lithium-ion batteries in the trash or regular recycling bins because they could start a fire. Recycle your device or battery at a safe battery recycling location. Visit call2recycle.org to find a recycling spot near you.

Charging an E-bike

Charge your battery in a flat, dry area away from children, direct sunlight, liquids, tripping hazards, and in a location where the e-bike is not at risk of falling.



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SECTION 1 Chemical product and company identification

Chemical Chinese name	: PETG (绿色, 黑色, 白色, 橙色, 蓝色, 蓝色 半透明, 红色, 红色 半透明, 黄色, 黄色 荧光, 绿色 半透明, 银, 灰色, 透明)
Chemical English name	: PETG (Green, Black, White, Orange, Blue, Blue Translucent, Red, Red Translucent, Yellow, Yellow Fluorescent, Green Translucent, Silver, Grey, Transparent)
Name of company	: UltiMaker
Title	: Supplier
Address	: The Netherlands Geldermalsen Watermolenweg 2
Zip code	: 4191 PN
Tel.	: +31 (0) 88 383 4000 (9 AM - 5 PM CET)
E-mail	: Product-Compliance@Ultimaker.com
Emergency number	: +31 (0) 88 383 4000 (during office hours: 9 AM - 5 PM CET)
Recommended use of the chemical	: 3D-Printer filament
Restricted use of the chemical	: This product must not be used in applications other than those identified above, without first seeking advice of the supplier

SECTION 2 Hazards identification

Emergency overview

Appearance: Filament, Various colours. Keep away from: Oxidizing agents. Non flammable. The product is non-reactive under normal conditions of use, storage and transport

To our knowledge, this product does not present any particular risk, provided it is handled in accordance with good occupational hygiene and safety practice

GHS hazard classification

Other hazards not mentioned above are Not applicable or No data is available.

Label elements

No data available

Physical and chemical hazards

No additional information available

Health hazards

Symptoms/effects : No acute and delayed symptoms and effects are observed

Symptoms/effects after skin contact : Risk of thermal burns on contact with molten product

Environmental hazards

No additional information available

Safety Data Sheet

PETG

Compiled according to GB/T 16483, GB/T 17519
Product Reference code:UM00007

Other hazards

Risk of thermal burns on contact with molten product

SECTION 3 Composition/information on ingredients

Product form : Mixture.

Ingredient(s)	Concentration or concentration ranges (w/w %)	CAS No.
Glycol-modified PET	98 – 100	25038-91-9
Titanium dioxide (Additive for PETG White, Green, Grey, Yellow, Blue)	< 1	13463-67-7
Solvent Yellow 114 (Additive for PETG Green, Yellow)	< 1	75216-45-4
Solvent Orange 60 (Additive for PETG Orange)	< 1	6925-69-5
Ethyl methacrylate (Additive for PETG White)	< 1	97-63-2
Carbon black (Additive for PETG Black, Grey, Green)	< 0.5	1333-86-4

SECTION 4 First-aid measures

Description of necessary first-aid measures

- First-aid measures general : If you feel unwell, seek medical advice (show the label where possible)
- First-aid measures after inhalation : Remove person to fresh air and keep comfortable for breathing.
In molten state: Hazardous vapours may be released
- First-aid measures after skin contact : Wash skin with plenty of water and soap.
Take off contaminated clothing.
In case of contact with molten product, cool rapidly with water and seek immediate medical attention. Do not attempt to remove molten product from skin because skin will tear easily.
Burns caused by molten material must be treated clinically
- First-aid measures after eye contact : Rinse eyes with water as a precaution.
In the event of contact with molten product: Immediately flush eyes thoroughly with water for at least 15 minutes.
Get immediate medical advice/attention.
- First-aid measures after ingestion : If you feel unwell, seek medical advice

Safety Data Sheet

PETG

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Product Reference code:UM00007

Most important symptoms/effects

- Symptoms/effects : No acute and delayed symptoms and effects are observed
- Symptoms/effects after skin contact : Risk of thermal burns on contact with molten product

Advices for first aid responders

No additional information available

Notes for the doctor

- Other medical advice or treatment : Treat symptomatically

SECTION 5 Fire-fighting measures

Extinguishing media

- Suitable extinguishing media : Use extinguishing media appropriate for surrounding fire: Water spray, Dry powder, Foam, Carbon dioxide
- Unsuitable extinguishing media : Do not use a solid water stream as it may scatter and spread fire

Specific hazards

- Hazardous decomposition products in case of fire : Under fire conditions, hazardous fumes will be present: Carbon dioxide, Carbon monoxide
- Explosion hazard : Material can accumulate some static charge during transfer
Prevent build-up of electrostatic charges (e.g. by grounding)

Advice for firefighters and protective measures

- Firefighting instructions : No additional information available
- Protection during firefighting : Do not attempt to take action without suitable protective equipment
Self-contained breathing apparatus
Complete protective clothing
- Precautionary measures fire : Do not allow run-off from fire-fighting to enter drains or water courses

SECTION 6 Accidental release measures

Personal precautions, protective equipment and emergency procedures

- Heat and ignition sources : Keep away from heat, sparks and flames
Keep out of direct sunlight
- General measures : No additional information available
- Personal Precautions, Protective Equipment and Emergency Procedures : No additional information available

For non-emergency personnel

- Protective equipment : Wear recommended personal protective equipment
Refer to section 8.2
Remove contaminated clothing and shoes
- Emergency procedures : None in particular
In molten state: Do not breathe vapours
Ventilate spillage area
Avoid contact with skin, eyes and clothing

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For emergency responders

Protective equipment : Do not attempt to take action without suitable protective equipment
For further information refer to section 8: "Exposure controls/personal protection"

Environmental precautions

Avoid release to the environment
Notify authorities if product enters sewers or public waters

Methods and material for containment and cleaning up

Methods for cleaning : No additional information available
Methods and Equipment for Containment and Cleaning up : Sweep up and put in a closed container for disposal
If melted: allow liquid to solidify before taking it up
For containment : No additional information available

Prevention measures for secondary accidents

Prevention Measures for Secondary Accidents : No additional information available
Other information : Dispose of materials or solid residues at an authorized site

SECTION 7 Handling and storage

Handling

Precautions for safe handling : Ensure good ventilation of the work station
In molten state: Do not breathe vapours
Avoid contact with skin, eyes and clothing
Wear personal protective equipment
Hygiene measures : Handle in accordance with good industrial hygiene and safety practice
Do not eat, drink or smoke when using this product.
Always wash hands after handling the product
Contaminated work clothing should not be allowed out of the workplace.
Wash contaminated clothing before reuse.
Local and general ventilation : No additional information available

Storage

Storage conditions : To guarantee the quality and properties of the product:
Store in a well-ventilated place.
Store in original container
Keep container tightly closed to avoid moisture absorption and contamination
Material used in packaging/containers : No additional information available
Incompatible materials : Oxidising agents.
Storage temperature : 0 – 30 °C (Relative air humidity: <50%)
Heat and ignition sources : Keep away from heat, sparks and flames
Keep out of direct sunlight

Safety Data Sheet

PETG

Compiled according to GB/T 16483, GB/T 17519
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SECTION 8 Exposure controls / Personal protection equipment

Occupational exposure limits

Carbon black (Additive for PETG Black, Grey, Green) (1333-86-4)	
China - Occupational Exposure Limits	
Local name	炭黑粉尘 # Carbon black dust
OEL TWA	4 mg/m ³ 总尘
Remark (CN)	G2B (对人可疑致癌 (Possibly carcinogenic to humans))
Regulatory reference	GBZ 2.1-2019
Titanium dioxide (Additive for PETG White, Green, Grey, Yellow, Blue) (13463-67-7)	
China - Occupational Exposure Limits	
Local name	二氧化钛粉尘 # Titanium dioxide dust
OEL TWA	8 mg/m ³ 总尘
OEL chemical category (CN)	Possibly carcinogenic to humans dust
Catalogue of Occupational Hazard Factors	Category 1 - Dusts
Remark (CN)	G2B (对人可疑致癌 (Possibly carcinogenic to humans))
Regulatory reference	GBZ 2.1-2019

Biological limit values

No additional information available

Monitoring methods

No additional information available

Appropriate engineering controls

Use process enclosures, local exhaust ventilation or other engineering controls to keep airborne levels below specified exposure limits. If user operations generate dust, fumes or mist, use ventilation to keep exposure to airborne particles below the exposure limit.

Ventilation conditions (1 printer): Provide a good standard of general ventilation, not less than 2 air changes per hour (assumes a room volume of: 30 m³)

Personal protective equipment

Thermal hazard protection : Risk of thermal burns on contact with molten product
Hazardous vapours may be released
In molten state: Wear respiratory protection/heat resistant gloves

Environmental exposure controls : Avoid release to the environment.

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Other information : Handle in accordance with good industrial hygiene and safety procedures
Do not eat, drink or smoke when using this product.
Wash hands immediately after handling the product
Take off contaminated clothing and wash before reuse.

Hand protection : None under normal conditions
Use insulated gloves when handling this material hot

Type	Material	Permeation	Thickness (mm)	Penetration	Standard
In molten state: , Chemically resistant protective gloves, Heat-resistant.	Nitrile rubber (NBR).	6 (> 480 minutes).	>0.35		EN 374, EN 407.

Eye protection : None under normal use
In molten state: Wear eye protection

Type	Use	Characteristics	Standard
Safety glasses with side shields.	In molten state.		EN 166.

Skin and body protection : None under normal use
In molten state: Wear suitable protective clothing

Type	Standard
Long sleeved protective clothing.	EN 13688.

Respiratory protection : None under normal use
In molten state: In case of insufficient ventilation, wear suitable respiratory equipment

Device	Filter type	Condition	Standard
Air-Purifying Respirator (APR), disposable.	Type B/P2.		EN 140, EN 14387.

SECTION 9 Physical and chemical properties

Physical state : Solid

Appearance : Filament

Colour : Various colours

Odour : Slight

pH : No data available

Melting point : 220 – 230 °C

Freezing point : Not applicable

Boiling point : Not applicable

Flash point : Not applicable

Auto-ignition temperature : Not applicable

Decomposition temperature : No data available

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PETG

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Flammability	: Non flammable
Vapour pressure	: No data available
Relative vapour density at 20°C	: No data available
Density	: 1.27 g/ml (25°C)
Solubility	: No data available
Solubility in water	: Insoluble
Partition coefficient n-octanol/water (Log Pow)	: No data available
Viscosity, kinematic	: Not applicable
Explosive limits (vol %)	: Not applicable
Lower explosive limit (LEL)	: No data available
Upper explosive limit (UEL)	: No data available
Radioactive	: No
Explosive properties	: Not explosive
Oxidising properties	: Non oxidizing
Particle size distribution	: Not applicable

SECTION 10 Stability and reactivity

Chemical stability	: Stable under normal conditions
Reactivity	: The product is non-reactive under normal conditions of use, storage and transport
Possibility of hazardous reactions	: No dangerous reactions known under normal conditions of use
Conditions to avoid	: None under recommended storage and handling conditions (see section 7). To avoid thermal decomposition, do not overheat
Incompatible materials	: Oxidising agents
Hazardous decomposition products	: Under normal conditions of storage and use, hazardous decomposition products should not be produced Under fire conditions, hazardous fumes will be present: Carbon dioxide, Carbon monoxide
Other properties	: No additional information available

SECTION 11 Toxicological information

Acute toxicity

Acute toxicity (oral)	: Not classified
Acute toxicity (dermal)	: Not classified
Acute toxicity (inhalation)	: Not classified

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Solvent Orange 60 (Additive for PETG Orange) (6925-69-5)	
LD50 dermal rat	> 2000 mg/kg bodyweight Animal: rat, Animal sex: female, Guideline: OECD Guideline 402 (Acute Dermal Toxicity)
Ethyl methacrylate (Additive for PETG White) (97-63-2)	
LD50 oral rat	13424 mg/kg bodyweight Animal: rat
LD50 dermal rabbit	> 9.1 g/kg (Source: NLM_HSDB)
LC50 Inhalation - Rat	55 mg/l air Animal: rat (OECD 403)

Skin corrosion/irritation

Skin corrosion/irritation : Not classified

Solvent Orange 60 (Additive for PETG Orange) (6925-69-5)	
pH	4.69 Temp.: 26,8 ° C Concentration: 1 vol%

Serious eye damage/eye irritation

Serious eye damage/irritation : Not classified

Solvent Orange 60 (Additive for PETG Orange) (6925-69-5)	
pH	4.69 Temp.: 26,8 ° C Concentration: 1 vol%

Respiratory or skin sensitisation

Respiratory or skin sensitisation : Not classified

Germ cell mutagenicity

Germ cell mutagenicity : Not classified

Carcinogenicity

Carcinogenicity : Not classified

Reproductive toxicity

Reproductive toxicity : Not classified

STOT - single exposure

STOT-single exposure : Not classified

Ethyl methacrylate (Additive for PETG White) (97-63-2)	
STOT-single exposure	May cause respiratory irritation.

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STOT - repeated exposure

STOT-repeated exposure : Not classified

Solvent Orange 60 (Additive for PETG Orange) (6925-69-5)	
NOAEL (oral, rat, 90 days)	500 mg/kg bodyweight Animal: rat, Guideline: OECD Guideline 407 (Repeated Dose 28-Day Oral Toxicity Study in Rodents)
Ethyl methacrylate (Additive for PETG White) (97-63-2)	
LOAEC (inhalation, rat, gas, 90 days)	350 ppm Animal: rat, Guideline: OECD Guideline 413 (Subchronic Inhalation Toxicity: 90-Day Study)

Aspiration hazard

Aspiration hazard : Not classified

PETG (Green, Black, White, Orange, Blue, Blue Translucent, Red, Red Translucent, Yellow, Yellow Fluorescent, Green Translucent, Silver, Grey, Transparent)	
Viscosity, kinematic	Not applicable
Density	1.27 g/ml (25°C)
Ethyl methacrylate (Additive for PETG White) (97-63-2)	
Viscosity, kinematic	0.7 mm² /s Temp.: '20° C' Parameter: 'kinematic viscosity (in mm² /s)'

SECTION 12 Ecological information

Ecotoxicity

Ecology - general : The product is not considered harmful to aquatic organisms nor to cause long-term adverse effects in the environment.

Hazardous to the aquatic environment, short-term (acute) : Not classified

Hazardous to the aquatic environment, long-term (chronic) : Not classified

Titanium dioxide (Additive for PETG White, Green, Grey, Yellow, Blue) (13463-67-7)	
LC50 fish 1	> 1000 mg/l
Solvent Yellow 114 (Additive for PETG Green, Yellow) (75216-45-4)	
Partition coefficient n-octanol/water (Log Kow)	4.8 @ 25 °C

Safety Data Sheet

PETG

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Product Reference code:UM00007

Solvent Orange 60 (Additive for PETG Orange) (6925-69-5)	
LC50 fish 1	> 0.5 µg/l Test organisms (species): Danio rerio (previous name: Brachydanio rerio)
EC50 72h - Algae [1]	> 2 mg/l Test organisms (species): Desmodesmus subspicatus (previous name: Scenedesmus subspicatus)
NOEC chronic fish	2.024 mg/l Test organisms (species): other: Duration: '28 d'
Partition coefficient n-octanol/water (Log Pow)	3.966 (at 25 °C (at pH 5.1)
Ethyl methacrylate (Additive for PETG White) (97-63-2)	
LC50 fish 1	100 mg/l Test organisms (species): Oncorhynchus mykiss
EC50 Daphnia 1	> 66 mg/l Test organisms (species): Daphnia magna
EC50 72h - Algae [1]	> 110 mg/l Test organisms (species): other

Persistence and degradability

PETG (Green, Black, White, Orange, Blue, Blue Translucent, Red, Red Translucent, Yellow, Yellow Fluorescent, Green Translucent, Silver, Grey, Transparent)	
Persistence and degradability	No additional information available
Solvent Orange 60 (Additive for PETG Orange) (6925-69-5)	
Not rapidly degradable	Yes

Bioaccumulative potential

Solvent Yellow 114 (Additive for PETG Green, Yellow) (75216-45-4)	
Partition coefficient n-octanol/water (Log Kow)	See section 12.1 on ecotoxicology4.8 @ 25 °C
Solvent Orange 60 (Additive for PETG Orange) (6925-69-5)	
Partition coefficient n-octanol/water (Log Pow)	3.966 (at 25 °C (at pH 5.1)

Mobility in soil

Solvent Yellow 114 (Additive for PETG Green, Yellow) (75216-45-4)	
Partition coefficient n-octanol/water (Log Kow)	See section 12.1 on ecotoxicology4.8 @ 25 °C

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Solvent Orange 60 (Additive for PETG Orange) (6925-69-5)	
Partition coefficient n-octanol/water (Log Pow)	3.966 (at 25 °C (at pH 5.1)

Other adverse effects

Classification procedure (Ozone) : No data available

SECTION 13 Disposal considerations

Waste treatment methods	: Dispose of contents/container in accordance with licensed collector's sorting instructions.
Contaminated container and packaging	: No additional information available
Additional information	: No additional information available
Product/Packaging disposal recommendations	: Empty containers should be taken for recycling, recovery or waste in accordance with local regulation
Regional waste regulation	: Dispose of in accordance with relevant local regulations

SECTION 14 Transport information

In accordance with JT/T 617 / IMDG / IATA

Overland transport (JT/T 617)	Transport by sea	Air transport
UN number		
Not regulated for transport		
Proper shipping name		
Not regulated	Not regulated	Not regulated
Transport hazard class(es)		
Not regulated	Not regulated	Not regulated
Packing group		
Not regulated	Not regulated	Not regulated
Environmental hazards		
Not regulated	Not regulated	Not regulated
No supplementary information available		

Special transport precautions

Overland transport (JT/T 617)

Not regulated

Transport by sea

Safety Data Sheet

PETG

Compiled according to GB/T 16483, GB/T 17519
Product Reference code:UM00007

Not regulated

Air transport

Not regulated

SECTION 15 Regulatory information

New Chemical Substance Environmental Management Registration Measures (MEE Order 12 of 2020)

Inventory of Existing Chemical Substances in China (IECSC) : Contains listed substance(s)
1,4-Benzenedicarboxylic acid, polymer with 1,4-cyclohexanedimethanol and 1,2-ethanediol (CAS-No. 25038-91-9)
Carbon black (CAS-No. 1333-86-4)
Titanium oxide (CAS-No. 13463-67-7)
12H-Phthaloperin-12-one (CAS-No. 6925-69-5)
Ethyl- α -methylacrylate (CAS-No. 97-63-2)

Regulations on the Safe Management of Hazardous Chemicals (Decree 591 of the State Council)

Catalogue of Hazardous Chemicals (2015) : Contains Hazardous Chemical(s)
Solvent Yellow 114
(Additive for PETG Green, Yellow)
Methacrylic acid ethyl ester, inhibitde (CAS-No. 97-63-2)

Law of the People's Republic of China on the Prevention and Control of Occupational Diseases

Catalogue for Classification of Hazardous Factors of Occupational Diseases : Contains listed substance(s)
Carbon black dust (CAS-No. 1333-86-4)
Titanium dioxide dust (CAS-No. 13463-67-7)

SECTION 16 Other information

SDS Reason for revision : Not applicable

Abbreviations and acronyms

CAS-No.	Chemical Abstract Service number
CAS	Chemical Abstract Service number
EN	European Standard
GHS	Globally Harmonized System of Classification and Labelling of Chemicals
IATA	International Air Transport Association
IMDG	International Maritime Dangerous Goods
OEL	Occupational Exposure Limit
SDS	Safety Data Sheet
TWA	Time Weighted Average

Training advice : Ensure staff are informed of and trained on the nature of exposure and basic actions to minimise exposure.

Safety Data Sheet

PETG

Compiled according to GB/T 16483, GB/T 17519
Product Reference code:UM00007

Section	Changed item	Change	Comments
1.	Chemical product and company identification.	Modified.	

Safety Data Sheet (SDS), China

DISCLAIMER OF LIABILITY The information in this SDS was obtained from sources which we believe are reliable. However, the information is provided without any warranty, express or implied, regarding its correctness. The conditions or methods of handling, storage, use or disposal of the product are beyond our control and may be beyond our knowledge. For this and other reasons, we do not assume responsibility and expressly disclaim liability for loss, damage or expense arising out of or in any way connected with the handling, storage, use or disposal of the product. This SDS was prepared and is to be used only for this product. If the product is used as a component in another product, this SDS information may not be applicable

PLA

Safety Data Sheet

according to Regulation (EC) No. 1907/2006 (REACH) with its amendment Regulation (EU) 2020/878
A safety data sheet is not required for this product. This document was created on a voluntary basis.
SDS ID: UM00008
Revision date: 06/12/2023 Supersedes version of: 22/12/2022 Version: 6.0

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Product form : Mixture
Trade name : PLA
(Green, Black, White, Transparent, Orange, Blue, Red, Yellow, Silver Metallic, Magenta, Pearl White)

1.2. Relevant identified uses of the substance or mixture and uses advised against

1.2.1. Relevant identified uses

Main use category : Professional use
Use of the substance/mixture : 3D-Printer filament

1.2.2. Uses advised against

Restrictions on use : This product must not be used in applications other than those identified above, without first seeking advice of the supplier

1.3. Details of the supplier of the safety data sheet

Supplier

UltiMaker
Watermolenweg 2
4191 PN Geldermalsen - The Netherlands
T +31 (0) 88 383 4000 (9 AM - 5 PM CET)
Product-Compliance@Ultimaker.com

1.4. Emergency telephone number

Country	Organisation/Company	Address	Emergency number	Comment
United Kingdom	National Health Service (NHS)		111 999 (in life-threatening emergencies)	
Wales	National Health Service (NHS)		0845 46 47	

SECTION 2: Hazards identification

2.1. Classification of the substance or mixture

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Not classified

Adverse physicochemical, human health and environmental effects

To our knowledge, this product does not present any particular risk, provided it is handled in accordance with good occupational hygiene and safety practice.

2.2. Label elements

Labelling according to Regulation (EC) No. 1272/2008 [CLP]

No labelling applicable

PLA

Safety Data Sheet

according to Regulation (EC) No. 1907/2006 (REACH) with its amendment Regulation (EU) 2020/878
A safety data sheet is not required for this product. This document was created on a voluntary basis.

2.3. Other hazards

Other hazards not contributing to the classification : Risk of thermal burns on contact with molten product.

The mixture does not contain substance(s) included in the list established in accordance with Article 59(1) of REACH for having endocrine disrupting properties, or is not identified as having endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at a concentration equal to or greater than 0,1 %

Other information : This substance/mixture does not meet the PBT/vPvB criteria of REACH regulation, annex XIII.

Component	
Polylactic acid (9051-89-2)	PBT: not relevant – no registration required vPvB: not relevant – no registration required
Titanium dioxide (Additive for PLA White, Green, Blue, Magenta) (13463-67-7)	This substance/mixture does not meet the PBT criteria of REACH regulation, annex XIII This substance/mixture does not meet the vPvB criteria of REACH regulation, annex XIII
Component	
Polylactic acid(9051-89-2)	The substance is not included in the list established in accordance with Article 59(1) of REACH for having endocrine disrupting properties, or is not identified as having endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605
Titanium dioxide (Additive for PLA White, Green, Blue, Magenta)(13463-67-7)	The substance is not included in the list established in accordance with Article 59(1) of REACH for having endocrine disrupting properties, or is not identified as having endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605

SECTION 3: Composition/information on ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Comments : Polylactic acid

Name	Product identifier	Conc. (% w/w)	Classification according to Regulation (EC) No. 1272/2008 [CLP]
Polylactic acid	(CAS-No.) 9051-89-2 (EC-No.) 618-575-7	> 95	Not classified
Carbon black (Additive for PLA Black)	(CAS-No.) 1333-86-4 (EC-No.) 215-609-9	< 2	Not classified
Titanium dioxide (Additive for PLA White, Green, Blue, Magenta)	(CAS-No.) 13463-67-7 (EC-No.) 236-675-5 (EC Index-No.) 022-006-00-2 (REACH-no) 01-2119489379-17	< 1	Not classified

Comments : Contains less than 1 % of titanium dioxide in the form of or incorporated in particles with aerodynamic diameter ≤ 10 µm

PLA

Safety Data Sheet

according to Regulation (EC) No. 1907/2006 (REACH) with its amendment Regulation (EU) 2020/878
A safety data sheet is not required for this product. This document was created on a voluntary basis.

SECTION 4: First aid measures

4.1. Description of first aid measures

First-aid measures general	: If you feel unwell, seek medical advice (show the label where possible).
First-aid measures after inhalation	: Remove person to fresh air and keep comfortable for breathing. In molten state: Hazardous vapours may be released.
First-aid measures after skin contact	: Take off contaminated clothing. Wash skin with plenty of water and soap. In case of contact with molten product, cool rapidly with water and seek immediate medical attention. Do not attempt to remove molten product from skin because skin will tear easily. Burns caused by molten material must be treated clinically.
First-aid measures after eye contact	: Rinse eyes with water as a precaution. In the event of contact with molten product: Immediately flush eyes thoroughly with water for at least 15 minutes. Get immediate medical advice/attention.
First-aid measures after ingestion	: Call a poison center or a doctor if you feel unwell.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms/effects	: No acute and delayed symptoms and effects are observed.
Symptoms/effects after skin contact	: Risk of thermal burns on contact with molten product.

4.3. Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

SECTION 5: Firefighting measures

5.1. Extinguishing media

Suitable extinguishing media	: Use extinguishing media appropriate for surrounding fire: Water spray, Dry powder, Foam, Carbon dioxide.
Unsuitable extinguishing media	: Do not use a solid water stream as it may scatter and spread fire.

5.2. Special hazards arising from the substance or mixture

Explosion hazard	: Material can accumulate some static charge during transfer. Prevent build-up of electrostatic charges (e.g. by grounding).
Hazardous decomposition products in case of fire	: Under fire conditions, hazardous fumes will be present: Carbon dioxide, Carbon monoxide, Aldehydes.

5.3. Advice for firefighters

Precautionary measures fire	: Do not allow run-off from fire-fighting to enter drains or water courses.
Protection during firefighting	: Do not attempt to take action without suitable protective equipment. Self-contained breathing apparatus. Complete protective clothing.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

6.1.1. For non-emergency personnel

Protective equipment	: Wear recommended personal protective equipment. Refer to section 8.2. Remove contaminated clothing and shoes.
Emergency procedures	: None in particular. In molten state: Do not breathe vapours. Ventilate spillage area. Avoid contact with skin, eyes and clothing.

6.1.2. For emergency responders

Protective equipment	: Do not attempt to take action without suitable protective equipment. For further information refer to section 8: "Exposure controls/personal protection".
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PLA

Safety Data Sheet

according to Regulation (EC) No. 1907/2006 (REACH) with its amendment Regulation (EU) 2020/878
A safety data sheet is not required for this product. This document was created on a voluntary basis.

6.2. Environmental precautions

Avoid release to the environment. Notify authorities if product enters sewers or public waters.

6.3. Methods and material for containment and cleaning up

Methods for cleaning up : Sweep up and put in a closed container for disposal. If melted: allow liquid to solidify before taking it up.
Other information : Dispose of materials or solid residues at an authorized site.

6.4. Reference to other sections

For further information refer to section 8: "Exposure controls/personal protection". For disposal of residues refer to section 13: "Disposal considerations".

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Precautions for safe handling : Ensure good ventilation of the work station. In molten state: Do not breathe vapours. Avoid contact with skin, eyes and clothing. Wear personal protective equipment.
Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Do not eat, drink or smoke when using this product. Always wash hands after handling the product. Contaminated work clothing should not be allowed out of the workplace. Wash contaminated clothing before reuse.

7.2. Conditions for safe storage, including any incompatibilities

Storage conditions : To guarantee the quality and properties of the product: Store in a well-ventilated place. Store in original container. Keep container tightly closed to avoid moisture absorption and contamination.
Incompatible materials : Oxidising agents. Strong bases.
Storage temperature : -20 – 30 °C (Relative air humidity: <50%)
Heat and ignition sources : Keep away from heat, sparks and flames. Keep out of direct sunlight.

7.3. Specific end use(s)

3D-Printer filament.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

8.1.1 National occupational exposure and biological limit values

Titanium dioxide (Additive for PLA White, Green, Blue, Magenta) (13463-67-7)

Ireland - Occupational Exposure Limits

Local name	Titanium dioxide
OEL (8 hours ref) (mg/m ³)	10 mg/m ³ total inhalable dust 4 mg/m ³ respirable dust
OEL (15 min ref) (mg/m ³)	30 mg/m ³ (calculated-respirable dust) 12 mg/m ³ (calculated)
Regulatory reference	Chemical Agents Code of Practice 2020

United Kingdom - Occupational Exposure Limits

Local name	Titanium dioxide
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PLA

Safety Data Sheet

according to Regulation (EC) No. 1907/2006 (REACH) with its amendment Regulation (EU) 2020/878
A safety data sheet is not required for this product. This document was created on a voluntary basis.

Titanium dioxide (Additive for PLA White, Green, Blue, Magenta) (13463-67-7)	
WEL TWA (mg/m ³)	4 mg/m ³ respirable 10 mg/m ³ total inhalable
WEL STEL (mg/m ³)	30 mg/m ³ (calculated-total inhalable) 12 mg/m ³ (calculated-respirable)
Regulatory reference	EH40/2005 (Fourth edition, 2020). HSE

Carbon black (Additive for PLA Black) (1333-86-4)	
Ireland - Occupational Exposure Limits	
Local name	Carbon black
OEL (8 hours ref) (mg/m ³)	3 mg/m ³ I (Inhalable Fraction)
Regulatory reference	Chemical Agents Code of Practice 2021
United Kingdom - Occupational Exposure Limits	
Local name	Carbon black
WEL TWA (mg/m ³)	3.5 mg/m ³
WEL STEL (mg/m ³)	7 mg/m ³
Regulatory reference	EH40/2005 (Fourth edition, 2020). HSE

8.1.2. Recommended monitoring procedures

Monitoring methods	
Monitoring methods	Refer to European Standard EN 689 (Workplace atmospheres - Guidance for the assessment of exposure by inhalation to chemical agents for comparison with limit values and measurement strategy) or equivalent national standard(s). Refer to European Standard EN 482 (Workplace atmospheres - General requirements for the performance of procedures for the measurement of chemical agents) or equivalent national standard(s). Refer to European Standard EN 14042 (Workplace atmospheres - Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents) or equivalent national standard(s).

8.1.3. Air contaminants formed

No additional information available

8.1.4. DNEL and PNEC

No additional information available

8.1.5. Control banding

No additional information available

8.2. Exposure controls

8.2.1. Appropriate engineering controls

Appropriate engineering controls:

Use process enclosures, local exhaust ventilation or other engineering controls to keep airborne levels below specified exposure limits. If user operations generate dust, fumes or mist, use ventilation to keep exposure to airborne particles below the exposure limit. Ventilation conditions (1 printer): Provide a good standard of general ventilation, not less than 2 air changes per hour (assumes a room volume of: 30 m³).

PLA

Safety Data Sheet

according to Regulation (EC) No. 1907/2006 (REACH) with its amendment Regulation (EU) 2020/878
A safety data sheet is not required for this product. This document was created on a voluntary basis.

8.2.2. Personal protection equipment

8.2.2.1. Eye and face protection

Eye protection:			
None under normal use. In molten state: Wear eye protection			
Type	Use	Characteristics	Standard
Safety glasses with side shields	In molten state		EN 166

8.2.2.2. Skin protection

Skin and body protection:	
None under normal use. In molten state: Wear suitable protective clothing	
Type	Standard
Long sleeved protective clothing	EN 13688

Hand protection:					
None under normal conditions. Use insulated gloves when handling this material hot					
Type	Material	Permeation	Thickness (mm)	Penetration	Standard
In molten state: Chemically resistant protective gloves, Heat-resistant	Nitrile rubber (NBR)	6 (> 480 minutes)	>0.35		EN 374, EN 407

8.2.2.3. Respiratory protection

Respiratory protection:			
None under normal use. In molten state: In case of insufficient ventilation, wear suitable respiratory equipment			
Device	Filter type	Condition	Standard
Air-Purifying Respirator (APR), disposable	Type B/P2		EN 140, EN 14387

8.2.2.4. Thermal hazards

Thermal hazard protection:

Risk of thermal burns on contact with molten product. Hazardous vapours may be released. In molten state: Wear respiratory protection/heat resistant gloves.

8.2.3. Other exposure controls

Environmental exposure controls:

Avoid release to the environment.

Other information:

Handle in accordance with good industrial hygiene and safety procedures. Do not eat, drink or smoke when using this product. Wash hands immediately after handling the product. Take off contaminated clothing and wash before reuse.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Physical state	: Solid
Colour	: Various colours
Appearance	: Filament
Odour	: Slight

PLA

Safety Data Sheet

according to Regulation (EC) No. 1907/2006 (REACH) with its amendment Regulation (EU) 2020/878
A safety data sheet is not required for this product. This document was created on a voluntary basis.

Odour threshold	: Not available
Melting point	: 145 – 160 °C
Freezing point	: Not applicable
Boiling point	: Not available
Flammability	: Non flammable
Explosive limits	: Not applicable
Flash point	: Not applicable
Auto-ignition temperature	: 388 °C
Decomposition temperature	: > 250 °C
pH	: Not available
Viscosity, kinematic	: Not applicable
Solubility	: Water: Insoluble Organic solvent: Chloroform
Vapour pressure	: Not available
Density	: 1.24 g/cm ³
Relative density	: Not available
Relative vapour density at 20°C	: Not applicable
Particle size	: Not available
Particle size distribution	: Not applicable

9.2. Other information

9.2.1. Information with regard to physical hazard classes

No additional information available

9.2.2. Other safety characteristics

No additional information available

SECTION 10: Stability and reactivity

10.1. Reactivity

The product is non-reactive under normal conditions of use, storage and transport.

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

No dangerous reactions known under normal conditions of use.

10.4. Conditions to avoid

None under recommended storage and handling conditions (see section 7). Do not expose to temperatures above 300 °C.

10.5. Incompatible materials

Oxidising agents. Strong bases.

10.6. Hazardous decomposition products

Under normal conditions of storage and use, hazardous decomposition products should not be produced. Under fire conditions, hazardous fumes will be present: Carbon dioxide, Carbon monoxide, Aldehydes.

SECTION 11: Toxicological information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity (oral)	: Based on available data, the classification criteria are not met
Acute toxicity (dermal)	: Based on available data, the classification criteria are not met

PLA

Safety Data Sheet

according to Regulation (EC) No. 1907/2006 (REACH) with its amendment Regulation (EU) 2020/878
A safety data sheet is not required for this product. This document was created on a voluntary basis.

Acute toxicity (inhalation)	: Based on available data, the classification criteria are not met
Skin corrosion/irritation	: Based on available data, the classification criteria are not met
Additional information	: Product dust may cause mechanical irritation to the skin and mucous membranes.
Serious eye damage/irritation	: Based on available data, the classification criteria are not met
Additional information	: Dust from this product may cause minor eye irritation
Respiratory or skin sensitisation	: Based on available data, the classification criteria are not met
Germ cell mutagenicity	: Based on available data, the classification criteria are not met
Carcinogenicity	: Based on available data, the classification criteria are not met
Reproductive toxicity	: Based on available data, the classification criteria are not met
STOT-single exposure	: Based on available data, the classification criteria are not met
STOT-repeated exposure	: Based on available data, the classification criteria are not met
Aspiration hazard	: Based on available data, the classification criteria are not met

PLA

(Green, Black, White, Transparent, Orange, Blue, Red, Yellow, Silver Metallic, Magenta, Pearl White)

Viscosity, kinematic	Not applicable
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11.2. Information on other hazards

11.2.1. Endocrine disrupting properties

Adverse health effects caused by endocrine disrupting properties	: Contains no substances identified as having endocrine disrupting properties
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11.2.2 Other information

No additional information available

SECTION 12: Ecological information

12.1. Toxicity

Ecology - general	: The product is not considered harmful to aquatic organisms nor to cause long-term adverse effects in the environment.
Hazardous to the aquatic environment, short-term (acute)	: Not classified
Hazardous to the aquatic environment, long-term (chronic)	: Not classified

Titanium dioxide

(Additive for PLA White, Green, Blue, Magenta) (13463-67-7)

LC50 fish 1	> 1000 mg/l
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12.2. Persistence and degradability

PLA

(Green, Black, White, Transparent, Orange, Blue, Red, Yellow, Silver Metallic, Magenta, Pearl White)

Persistence and degradability	Not biodegradable.
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Polylactic acid (9051-89-2)

Biodegradation	Not biodegradable
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12.3. Bioaccumulative potential

PLA

(Green, Black, White, Transparent, Orange, Blue, Red, Yellow, Silver Metallic, Magenta, Pearl White)

Bioaccumulative potential	No bioaccumulation.
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PLA

Safety Data Sheet

according to Regulation (EC) No. 1907/2006 (REACH) with its amendment Regulation (EU) 2020/878
A safety data sheet is not required for this product. This document was created on a voluntary basis.

12.4. Mobility in soil

No additional information available

12.5. Results of PBT and vPvB assessment

Component	
Polylactic acid (9051-89-2)	PBT: not relevant – no registration required vPvB: not relevant – no registration required
Titanium dioxide (Additive for PLA White, Green, Blue, Magenta) (13463-67-7)	This substance/mixture does not meet the PBT criteria of REACH regulation, annex XIII This substance/mixture does not meet the vPvB criteria of REACH regulation, annex XIII

12.6. Endocrine disrupting properties

Adverse effects on the environment caused by endocrine disrupting properties : Contains no substances identified as having endocrine disrupting properties

12.7. Other adverse effects

No additional information available

SECTION 13: Disposal considerations

13.1. Waste treatment methods

Regional waste regulation : Dispose of in accordance with relevant local regulations.
Waste treatment methods : Dispose of contents/container in accordance with licensed collector's sorting instructions.
Product/Packaging disposal recommendations : Empty containers should be taken for recycling, recovery or waste in accordance with local regulation.

SECTION 14: Transport information

In accordance with ADR / IMDG / IATA / ADN / RID

ADR	IMDG	IATA	ADN	RID
14.1. UN number or ID number				
Not regulated	Not regulated	Not regulated	Not regulated	Not regulated
14.2. UN proper shipping name				
Not regulated	Not regulated	Not regulated	Not regulated	Not regulated
14.3. Transport hazard class(es)				
Not regulated	Not regulated	Not regulated	Not regulated	Not regulated
14.4. Packing group				
Not regulated	Not regulated	Not regulated	Not regulated	Not regulated
14.5. Environmental hazards				
Not regulated	Not regulated	Not regulated	Not regulated	Not regulated
No supplementary information available				

PLA

Safety Data Sheet

according to Regulation (EC) No. 1907/2006 (REACH) with its amendment Regulation (EU) 2020/878
A safety data sheet is not required for this product. This document was created on a voluntary basis.

14.6. Special precautions for user

Overland transport

Not regulated

Transport by sea

Not regulated

Air transport

Not regulated

Inland waterway transport

Not regulated

Rail transport

Not regulated

14.7. Maritime transport in bulk according to IMO instruments

Not applicable

SECTION 15: Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

15.1.1. EU-Regulations

Contains no substance(s) listed on REACH Annex XVII (Restriction Conditions)

Contains no substance(s) listed on the REACH Candidate List

Contains no substance(s) listed on REACH Annex XIV (Authorisation List)

Contains no substance(s) listed on the PIC list (Regulation EU 649/2012 concerning the export and import of hazardous chemicals)

Contains no substance(s) listed on the POP list (Regulation EU 2019/1021 on persistent organic pollutants)

15.1.2. National regulations

No additional information available

15.2. Chemical safety assessment

Not required

No chemical safety assessment has been carried out

SECTION 16: Other information

Indication of changes:			
Section	Changed item	Change	Comments
3.1	Composition/information on ingredients	Modified	

Training advice : Ensure staff are informed of and trained on the nature of exposure and basic actions to minimise exposure.

Abbreviations and acronyms:	
ADN	European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways
ADR	European Agreement concerning the International Carriage of Dangerous Goods by Road
CAS	Chemical Abstract Service number
CLP	Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008
IATA	International Air Transport Association
IMDG	International Maritime Dangerous Goods

PLA

Safety Data Sheet

according to Regulation (EC) No. 1907/2006 (REACH) with its amendment Regulation (EU) 2020/878
A safety data sheet is not required for this product. This document was created on a voluntary basis.

REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals Regulation (EC) No 1907/2006
RID	Regulations concerning the International Carriage of Dangerous Goods by Rail
vPvB	Very Persistent and Very Bioaccumulative
PBT	Persistent Bioaccumulative Toxic
SDS	Safety Data Sheet

Safety Data Sheet applicable for regions : IE - Ireland;GB - United Kingdom

SDS EU (CLP) - UM

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.